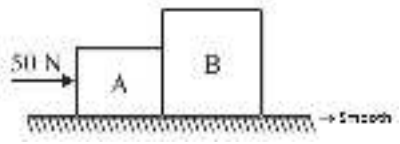
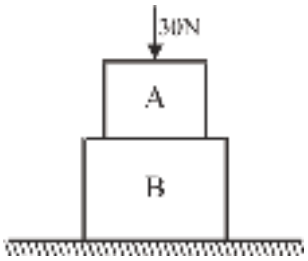
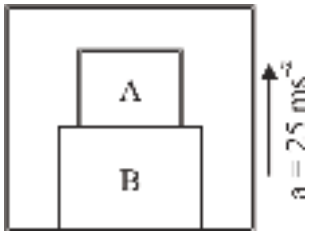
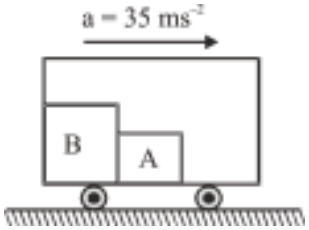
PHYSICS

1) In which of the following cases the contact force between A & B is maximum?
($m_A = m_B = 1kg$)

(1) 

(2) 

(3) 

(4) 

2) A body of mass 4 kg experiences two forces $\vec{F}_1 = 5\hat{i} + 8\hat{j} + 7\hat{k}$ and $\vec{F}_2 = 3\hat{i} - 4\hat{j} - 3\hat{k}$. The acceleration of the body is :

(1) $-2\hat{i} - \hat{j} - \hat{k}$

(2) $4\hat{i} + 2\hat{j} + 2\hat{k}$

(3) $2\hat{i} + \hat{j} + \hat{k}$

(4) $2\hat{i} + 3\hat{j} + 3\hat{k}$

3) A long conveyor belt is moving with a constant velocity of 10 m/s. A block is gently placed on the belt. If the coefficient of static and kinetic friction between the block and the belt is 0.5, find the distance travelled by the block relative to the ground before it stops slipping and starts moving with the belt. (Take g = 10 m/s^2)

(1) 5 m

(2) 10 m

(3) 20 m

(4) 25 m

4) The linear momentum of a particle varies with time t as $p = a + bt + ct^2$. Which of the following statement is correct :-

(1) Force varies with time in a quadratic manner

(2) Force is time dependent

(3) The velocity of the particle is proportional to time

(4) The displacement of the particle is proportional to time

5) A particle of mass m is moving in a horizontal circle of radius R with a constant speed v. If the particle rotates through an angle θ at the centre, determine the magnitude of the average force acting on the particle during this interval.

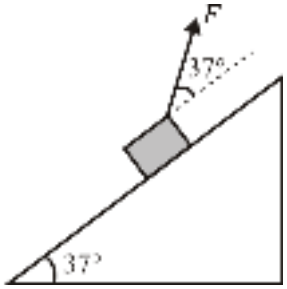
(1) mv^2 / R

(2) $\left(\frac{mv^2}{R}\right)\left[\frac{\sin(\theta/2)}{(\theta/2)}\right]$

(3) $(mv^2 / R)\sin(\theta)$

(4) Zero

6) A block resting on a smooth inclined plane is acted upon by a force F as shown. If mass of block is 2 kg and F = 20 N and sin 37°= 3/5, the acceleration of block is



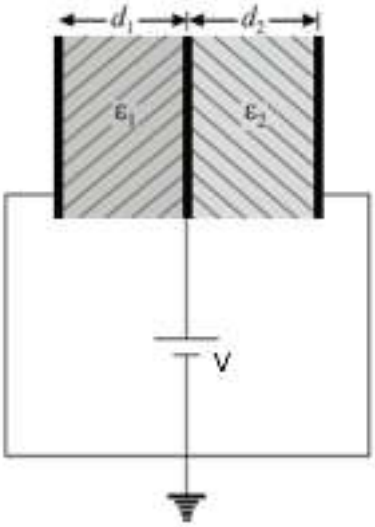
(1) 2 m/s^2

(2) 6 m/s^2

(3) 8 m/s^2

(4) zero

7) A capacitor is composed of three parallel conducting plates. All three plates are of same area A. The first pair of plates are kept a distance d_1 apart and the space between them is filled with a medium of a dielectric ε_1. The corresponding values for the second pair of plates are d_2 and ε_2



respectively. What is the surface charge density on the middle plate?

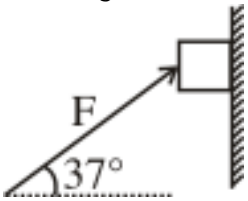
(1) $\varepsilon_0 V\left[\frac{\varepsilon_1}{d_1}+\frac{\varepsilon_2}{d_2}\right]$

(2) $-\varepsilon_0 V\left[\frac{\varepsilon_1}{d_1}+\frac{\varepsilon_2}{d_2}\right]$

(3) $2\varepsilon_0 V\left[\frac{\varepsilon_1}{d_1}+\frac{\varepsilon_2}{d_2}\right]$

(4) $-2\varepsilon_0 V\left[\frac{\varepsilon_1}{d_1}+\frac{\varepsilon_2}{d_2}\right]$

8) A 1 kg block is being pushed against a wall by a force F = 75 N as shown in the figure. The coefficient of friction is 0.25. The magnitude of acceleration of the block is :-



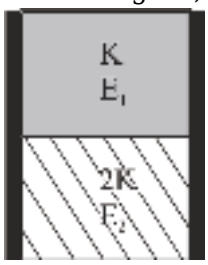
(1) 10 m/s^2

(2) 20 m/s^2

(3) 5 m/s^2

(4) None

9) A parallel plate capacitor is connected from a cell and then isolated from it. Two dielectric slabs of dielectric constant K and 2K are now introduce in the region between upper half and lower half of the plate (as shown in figure). The electric field intensity in upper half of dielectric is E_1 and lower half is E_2 then



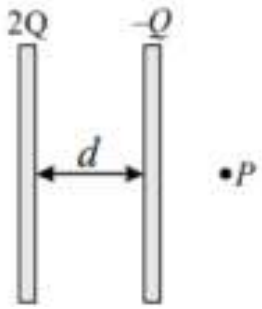
(1) $E_1 = 2E_2$

(2) Electrostatic potential energy of upper half is less than that of lower half

(3) Induced charges on both slabs are same

(4) Charge distribution on the plates remains same after insertion of dielectric

10) In the figure shown, the plates of a parallel plate capacitor have unequal charges. Its capacitance is C and P is a point outside the capacitor as shown in figure. The distance between the plates is d. Select the INCORRECT statement : -



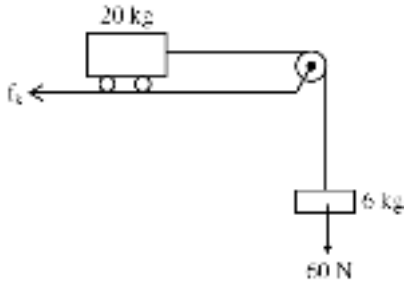
(1) A point charge at point P will experience an electric force due to capacitor.

(2) The potential difference between the plates will be 3Q/2C

(3) The energy stored in the electric field in the region between the plates is $\frac{9Q^2}{8C}$.

(4) The force on one plate due to the other plate is $\frac{Q^2}{2\pi\varepsilon_0 d^2}$.

11) Consider a block and trolley system as shown in figure. If the coefficient of kinetic friction between the trolley and the surface is 0.04, the acceleration of the system in ms^{-2} is :
(Consider that the string is massless and unstretchable and the pulley is also massless and frictionless) :



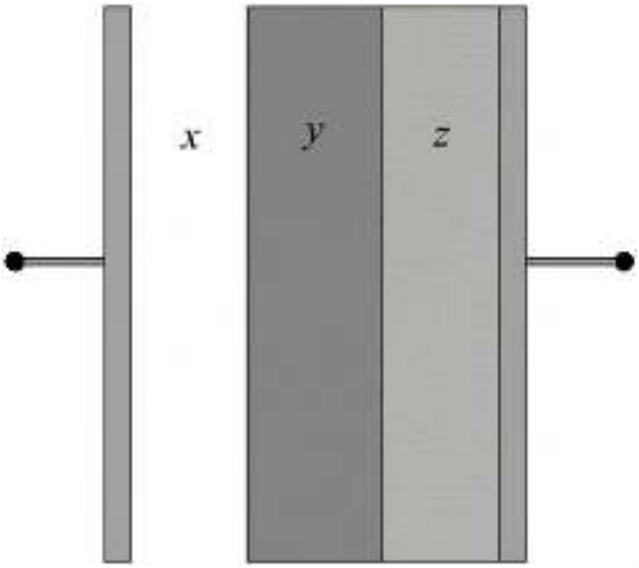
(1) 3

(2) 4

(3) 2

(4) 1.2

12) The figure shows a capacitor having three dielectric layers parallel to its plates. Layer x is vacuum, y is conductor and z is a dielectric. Which of the following change(s) will result in increase

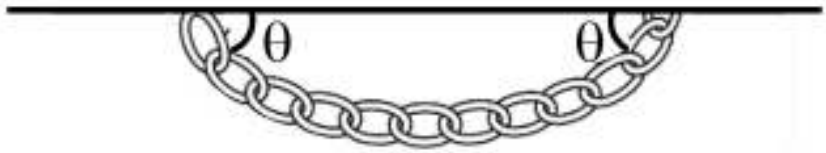


in capacitance?

(A) Replace x by conductor
(B) Replace y by dielectric
(C) Replace z by conductor
(D) Replace x by dielectric

(1) A,B,C
(2) A,C,D
(3) B,C,D
(4) A,B,C,D

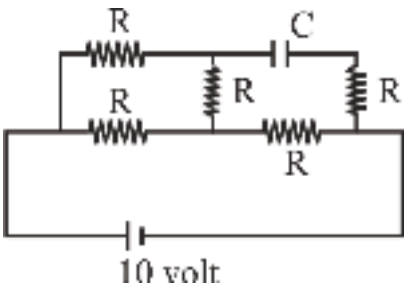
13) A uniform flexible chain of total weight **W** hangs between two supports at the same level. If the chain makes an angle **θ** with the horizontal at each support, what is the tension at the midpoint?



(1) W / 2
(2) W / (2 sin θ)
(3) (W/2) tan θ
(4) (W/2) cot θ

14) A capacitor stores 50 μC charge when connected across a battery. When the gap between the plates is filled with a dielectric, a charge of 100 μC flows through the battery. The dielectric constant of the material is : -

(1) 2.5
(2) 2
(3) 4
(4) 3



15) Find the potential difference across the capacitor in volts :-

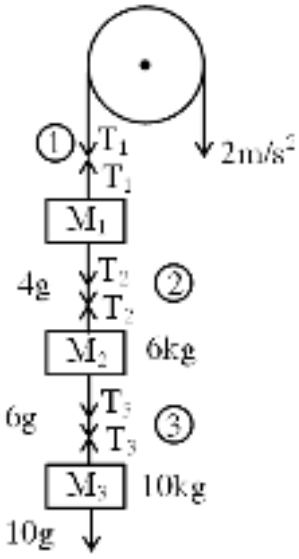
(1) 2 V

(2) 8 V

(3) 4 V

(4) 1 V

16) Three blocks M_1, M_2, M_3 having masses 4 kg, 6 kg and 10 kg respectively are hanging from a smooth pully using rope 1, 2 and 3 as shown in figure. The tension in the rope 1, T_1 when they are moving upward with acceleration of $2ms^{-2}$ is ..... N (if g = 10 m/s^2)



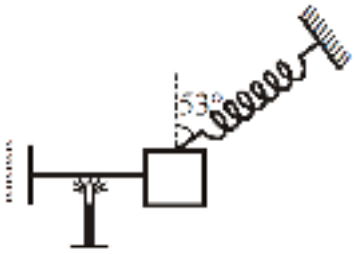
(1) 120 N

(2) 240 N

(3) 360 N

(4) 420 N

17) The block shown in the figure is in equilibrium. Find acceleration of the block just after the string burns.



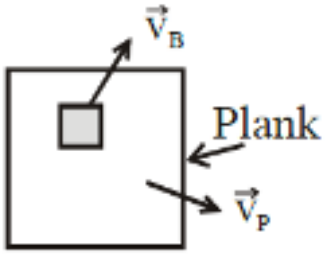
(1) $\frac{3g}{5}$

(2) $\frac{4g}{5}$

(3) $\frac{4g}{3}$

(4) None of these

18) Find the unit vector in the direction of the friction force acting on the block. The velocity of the platform is $\vec{v}_P = 3\hat{i} - 4\hat{j}$ and the velocity of the block is $\vec{v}_B = 7\hat{i} - 7\hat{j}$.



(1) $-\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j}$

(2) $\frac{4}{5}\hat{i} - \frac{3}{5}\hat{j}$

(3) $\frac{-4\hat{i} - 3\hat{j}}{5}$

(4) $\frac{4\hat{i} + 3\hat{j}}{5}$

19) In an air collision between an aeroplane and a bird, the force experienced by the bird as compared to that of the aeroplane is :-

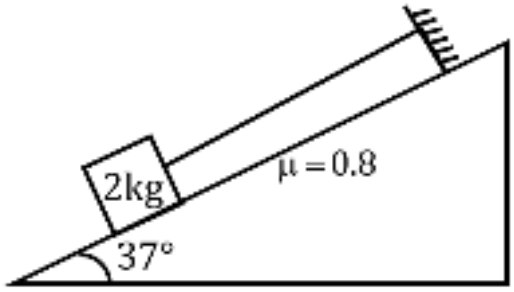
(1) Very high

(2) Equal

(3) Less

(4) Zero

20) A 2 kg block is placed on a rough wedge with an inclination of 37 degrees. The block is tied to the top edge of the wedge by a string. If the coefficient of static friction between the block and the wedge is 0.8, calculate the tension in the string?



(1) 20 N

(2) 12 N

(3) 0.8 N

(4) 0 N

21) Two identical capacitors are connected in series with a source of constant voltage V. If Q is the

charge on one of the capacitors, the capacitance of each capacitor is : -

(1) Q/2V

(2) Q/V

(3) 2Q/V

(4) None of these

22) Capacitor C_1 of the capacitance 1μF and another capacitor C_2 of capacitance 2μF are separately charged fully by a common battery. The two capacitors are then separately allowed to discharge through equal resistors at time t = 0.
(A) The current in each of the two discharging circuits is zero at t = 0.
(B) The current in the two discharging circuits at t = 0 are equal but not zero.
(C) The currents in the two discharging circuits at t = 0 are unequal.
(D) Capacitor C_1 loses 50% of its initial charge sooner than C_2 loses 50% of its initial charge.

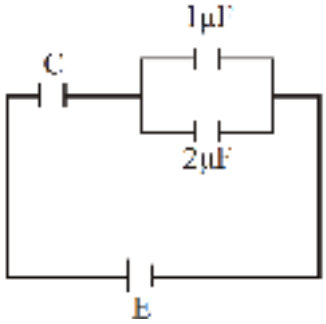
(1) None of the above

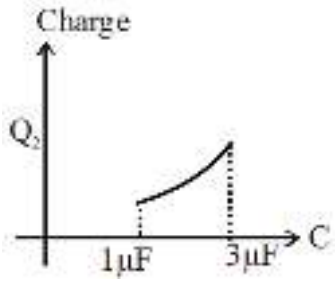
(2) B,D

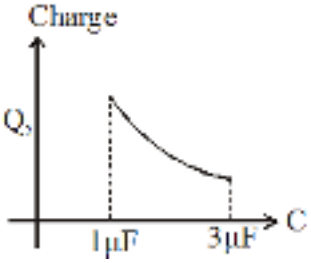
(3) C,D

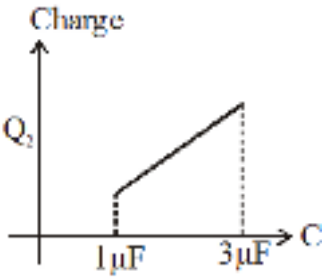
(4) A,D

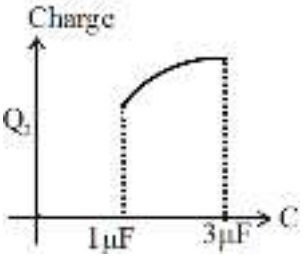
23) In the given circuit, charge Q_2 on the 2μF capacitor changes as C is varied from 1μF to 3μF. Q_2 as a function of 'C' is given properly by : (figures are drawn schematically and are not to scale) :-

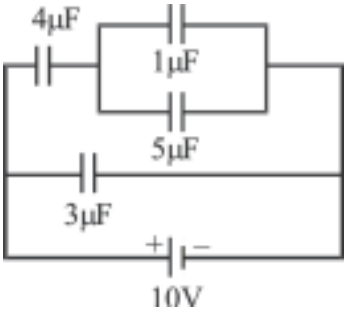


(1)


(2)


(3) 

(4) 



24) For the circuit shown in the figure, the charge on 4μF capacitor is :-

(1) 30μC

(2) 40μC

(3) 24μC

(4) 54μC

25) Capacity of an isolated sphere is increased n times when it is enclosed by an earthed concentric sphere. The ratio of their radii is :

(1) $\frac{n^2}{n-1}$

(2) $\frac{n}{n-1}$

(3) $\frac{2n}{n+1}$

(4) $\frac{2n+1}{n+1}$

26) Two persons A and B are sitting in two cars 1 and 2 such that the velocity of the cars is given by $\vec{v}_1 = 3t^2\hat{i} - \frac{4}{3}t^6\hat{j}$ and $\vec{v}_2 = 3\left(t^2 - 5\right)\hat{i}$ respectively. The fictitious (pseudo) force on B as observed by A at t = 1 sec will be :-
(m_A = 70 kg, m_B = 50kg)

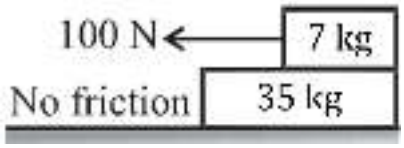
(1) $70\left(4\hat{j} - 3\hat{i}\right)$

(2) $100\left(4\hat{j}-3\hat{i}\right)$

(3) $50\left(8\hat{j}-15\hat{i}\right)$

(4) $70\left(8\hat{j}-15\hat{i}\right)$

27) A 35 kg slab (B) rests on a smooth floor as shown in figure. A 7 kg block (A) rests on the top of the slab. The static coefficient of friction between slab and block is 0.5 while the kinetic friction coefficient is 0.4. The block (A) is acted upon by a horizontal force 100 N. If g = 10 m/s^2, the resulting acceleration of the slab (B) will be :-



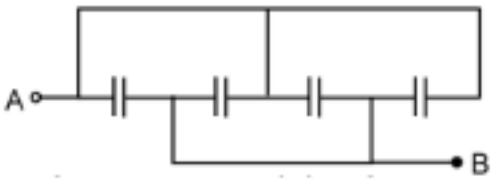
(1) 0.98 m/s^2

(2) 1.47 m/s^2

(3) 0.8 m/s^2

(4) 6.1 m/s^2

28) Four condensers are joined as shown in the figure. The capacitance of each capacitor is 8 μF. The equivalent capacitance between the points A and B will be : -



(1) 32 μF

(2) 2 μF

(3) 8 μF

(4) 16 μF

29) Capacitor A has a capacitance 15 μF when it is filled with a medium of dielectric constant 15. Another capacitor B has a capacitance 1 μF with air between the plates. Both are charged separately by a battery of 100 V. After charging, both are connected in parallel without the battery and the dielectric material is removed. The common potential now is : -

(1) 400 V

(2) 800 V

(3) 1200 V

(4) 1600 V

30) A capacitor is charged by using a battery, which is then disconnected. A dielectric slab is then inserted between the plates, which results in : -

(1) Reduction of charge on the plates and increase of potential difference across the plates.

(2) Increase in the potential difference across the plates, reduction in stored energy, but no change in the charge on the plates.

(3) Decrease in the potential difference across the plates, reduction in stored energy, but no change in the charge on the plates.

(4) None of the above.

31) A body of mass 5 kg starts from the origin with an initial velocity $\vec{u} = (80\hat{i} + 40\hat{j})ms^{-1}$. If a constant force $(-12\hat{i} - 5\hat{j})N$ acts on the body, the time in which the y component of the velocity becomes zero, is

(1) 5s

(2) 20s

(3) 40s

(4) 80s

32) Rope of length 10 m and linear mass density 0.5 kg/m is lying lengthwise on a smooth horizontal floor. It is pulled by a force of 25 N. Tension in the rope at a point 8 m away from the point of application of force is :-

(1) 20 N

(2) 15 N

(3) 10 N

(4) 5 N

33) A body moves with velocity $v = \ell nx$ where x is its position. The net force acting on body is zero at :

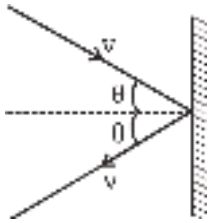
(1) 0 m

(2) $x = e^2$ m

(3) x = e m

(4) x = 1m

34) A water jet, whose cross sectional area is $2cm^2$ strikes a wall making an angle 60° with the normal and rebounds elastically. The velocity of water of density 1000 kg/m^3 is 3m/s. Force exerted on wall is :-



(1) 1.8N

(2) 1.2N

(3) 0.9N

(4) 3.6N

35) Three blocks of mass 1kg, 4kg and 2kg are placed on a smooth horizontal plane as shown in the figure. The contact force between 1kg block and 4 kg block is :-



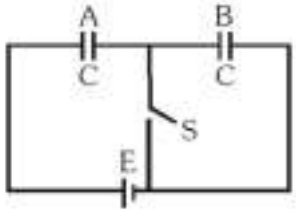
(1) 100 N

(2) 120 N

(3) 50 N

(4) 110N

36) Consider the situation shown. The switch S is open for a long time and then closed. Then :



	List-I		**List-II**
(P)	Charge flown through battery after S is closed	(1)	$\frac{CE^2}{2}$
(Q)	Work done by battery	(2)	$\frac{CE}{2}$
(R)	Charge on capacitor A long after switch S is closed	(3)	$\frac{CE^2}{4}$
(S)	Heat developed in the system	(4)	CE

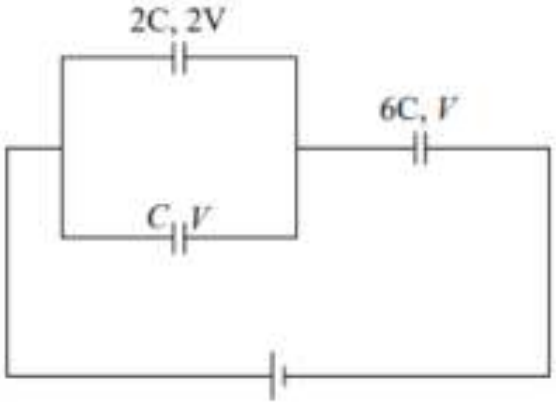
(1) P→2; Q→2; R→4; S→3

(2) P→2; Q→2; R→4; S→1

(3) P→1; Q→2; R→4; S→3

(4) P→ 2; Q→1; R→4; S→3

37) Circuit in figure shows three capacitors with capacitance and their breakdown voltage. What should be maximum value of the external source voltage such that no capacitor breaks down?



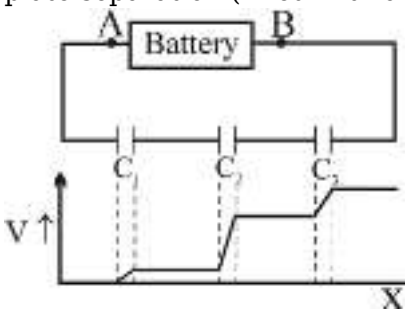
(1) 1V

(2) 2V

(3) 1.5V

(4) 4V

38) Figure shows a circuit containing a battery and three parallel plate capacitors with identical plate separation (filled with air). The variation of potential along the path A to B is shown below.



If the area of the plates of capacitors are A_1, A_2 & A_3 respectively then

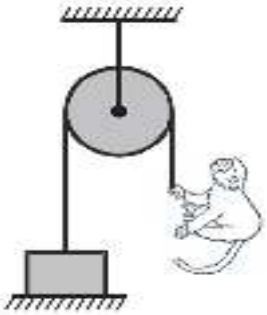
(1) $A_1 < A_2 > A_3$

(2) $A_1 > A_3 > A_2$

(3) $A_1 = A_2 = A_3$

(4) $A_1 > A_2 > A_3$

39) A monkey weighing 10 kg is climbing up a light rope which passes over an ideal pulley. The other end of the rope is attached a 15 kg mass as shown in the figure. In order to raise the 15 kg



mass off the ground the monkey should climb up :

(1) with constant acceleration g/3

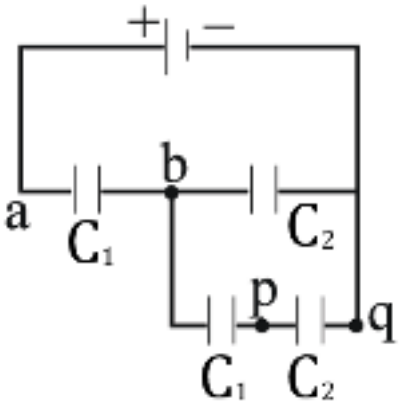
(2) with an acceleration greater than g/2

(3) with an acceleration greater than g/4

(4) It is not possible because weight of monkey is lesser than the block

40)

In the given network if potential difference between p and q is 2V and $C_2 = 3C_1$. Then if the potential difference between a & b is N volt. Calculate $\left(\frac{N+100}{100}\right)$.



(1) 1.30

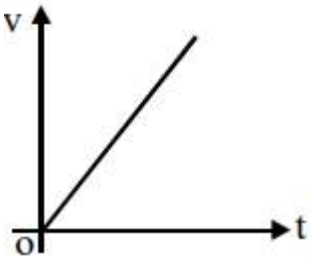
(2) 2.60

(3) 3.60

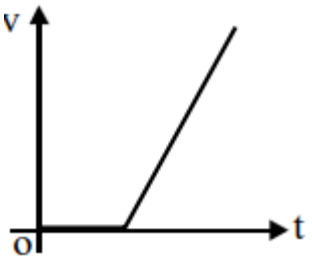
(4) 1.00

41) A metal block is resting on a rough wooden surface. A horizontal force applied to the block is increased uniformly. Which of the following curves correctly represents velocity of the block with respect to time?

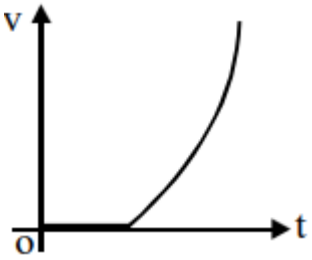
(1)



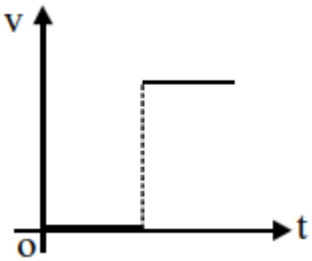
(2)



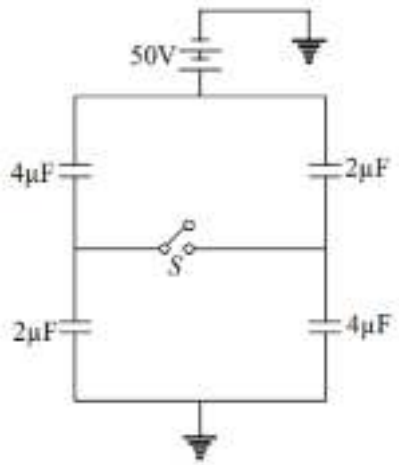
(3)



(4)



42) The circuit shown in figure below is in steady state. Now the switch S is closed. Calculate the



charge that flows through the switch is : -

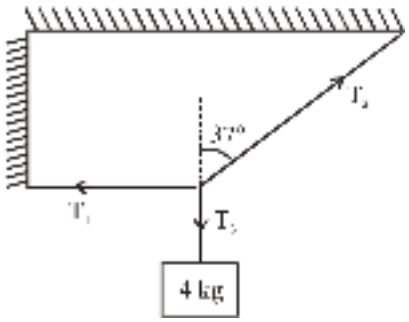
(1) $\frac{400}{3}$ μC

(2) 100 μC

(3) 50 μC

(4) $\frac{100}{3}$ μC

43) Calculate the tensions T_1, T_2 and T_3 in the three threads shown in the following figure. (All threads are mass less) (g = 10 m/s^2) :-



(1) 30 N, 40 N, 50 N

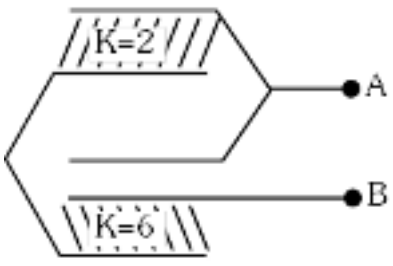
(2) 50 N, 30 N, 40 N

(3) 35 N, 45 N, 40 N

(4) 30 N, 50 N, 40 N

44)

Find equivalent capacitance between A and B ? (Area of each plate = A, separation between adjacent plate = d)



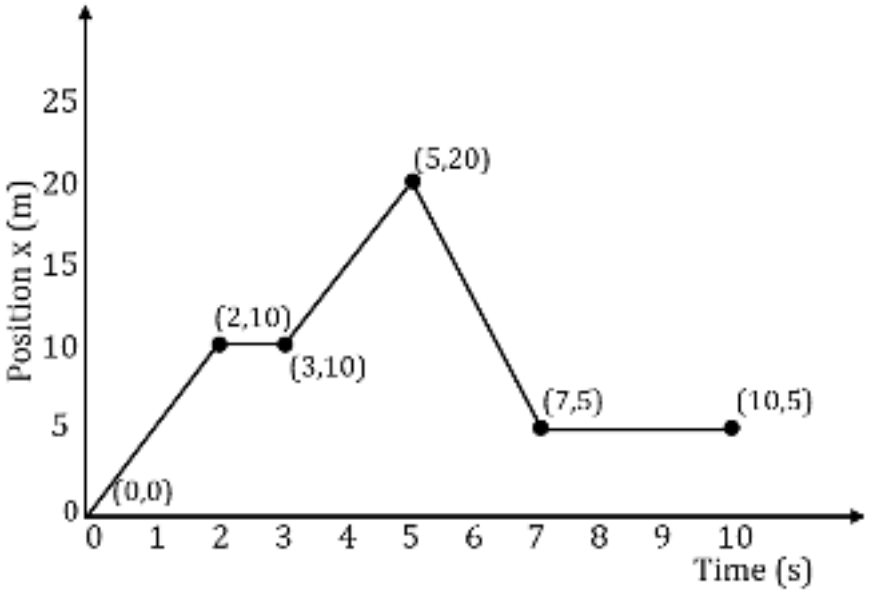
(1) $\frac{4A\varepsilon_0}{d}$

(2) $\frac{3A\varepsilon_0}{d}$

(3) $\frac{5A\varepsilon_0}{2d}$

(4) $\frac{2A\varepsilon_0}{d}$

45) A body of mass **1 kg** is moving along a straight line. Its displacement-time (x-t) graph is provided. Based on the graph, calculate the magnitude of the **impulse** imparted to the body at **t = 5 seconds**.



(1) 5.0 N-s

(2) 12.5 N-s

(3) 7.5 N-s

(4) 15.0 N-s

CHEMISTRY

1) Determine bond length order?
α. CH_3 – CH_3
β. CH_2 = CH_2
γ. CH_2 = CH – CH = CH_2

(1) α > γ > β

(2) β > α > γ

(3) α = β = γ

(4) none of these

2) Which of the following is correct order of acidic strength ?

(1) H_2O_2 > H_2O

(2) $H_3PO_3 < H_2SO_4$

(3) $H_2S < HBr$

(4) All of the above

3) Which compound exist :

(1) OF_4

(2) SCl_6

(3) MnF_7

(4) IF_7

4) Which of the following is incorrect order of boiling point.

(1) $PH_3 < AsH_3 < NH_3 < SbH_3$

(2) $H_2S < H_2Se < H_2Te < H_2O$

(3) $HCl < HBr < HI < HF$

(4) $SiH_4 < GeH_4 < CH_4 < SnH_4$

5) Consider the molecules given below:
(A) SF_4 (B) BF_3
(C) XeF_4 (D) SO_3
Select correct statement.

(1) (B) and (D) are polar molecules

(2) (A) is most polar molecule among the above given molecules

(3) (B) is most polar molecule among the above given molecules

(4) (A) and (C) are non-polar molecules

6) The boiling point of water is exceptionally high because :

(1) Water molecule is linear

(2) Water molecule is not linear

(3) There is covalent bond between H and O

(4) Water molecules associate due to hydrogen bonding

7) **Assertion :** In $AB_3 L_2$ molecule both lone pairs are present at axial position
Reason : Both lone pair on axial position in $AB_3 L_2$ minimize overall valence shell electron pair repulsion.

(1) Both Assertion and Reason are correct and Reason is correct for the Assertion.

(2) Both Assertion and Reason are correct but Reason is not correct for Assertion

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

8) Which statements are true about noble gases :
(A) The atomic radii values are very large.

(B) They are diatomic
(C) Radii of noble gases should be compared with Vander Waals radii not with covalent radii

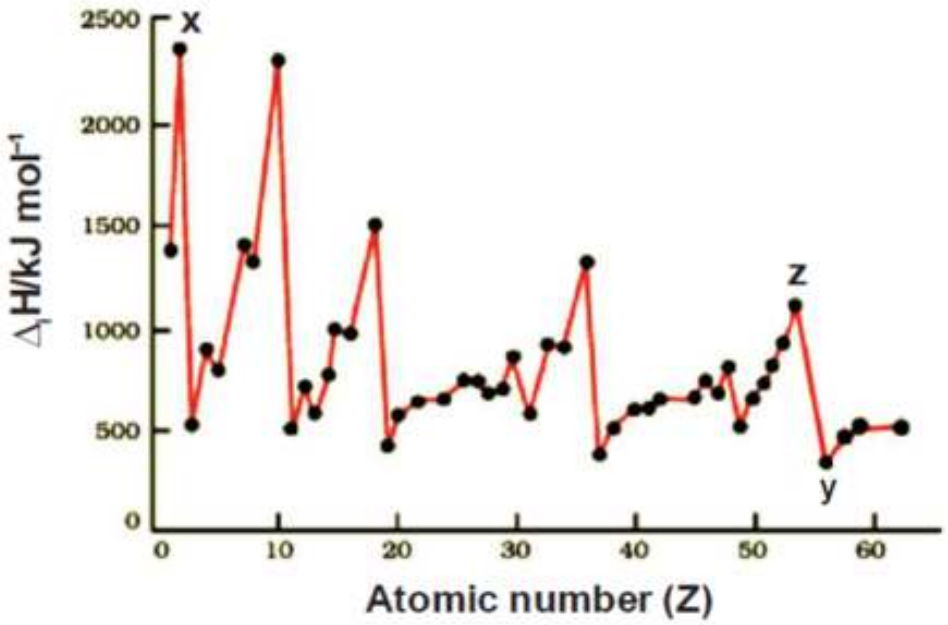
(1) Only A

(2) Both A and B

(3) Both A and C

(4) Both B and C

9) In the given graph of ionisation enthalpy. Find out the element x, y and z respectively :



(1) Ne, Cs and Xe

(2) Ne, Fr and Rn

(3) He, Cs and Xe

(4) He, Fr and Xe

10) Greatest amount of energy involves in the transformation of :

(1) $Be \rightarrow Be^{+} + e^{-}$

(2) $Mg \rightarrow Mg^{+} + e^{-}$

(3) $Li^{+} \rightarrow Li^{+2} + e^{-}$

(4) $He \rightarrow He^{+} + e^{-}$

11) Non metals are usually solids or gases at room temperature with low melting point and boiling point, the exceptions are :
(A) Carbon
(B) Boron
(C) Bromine
Choose the correct answer from the option given below :

(1) A, B only

(2) B, C only

(3) A, C only

(4) A, B and C

12) The order in which the energy of the available orbitals in 5th period are :

(1) 5s < 5p < 5d

(2) 5s < 4d < 5p

(3) 4s < 3d < 4p

(4) 4s < 4p

13) How many elements of 4th period will have one or more electrons in their 4d subshell :

(1) Ten

(2) Eight

(3) Eighteen

(4) Zero

14) Select the statement from the given and choose the initials as (T) for true and (F) for false :
(A) In p-block only non metals are present
(B) 1st electron gain enthalpy will always be exothermic
(C) As positive charge on cation increases, z-effective increases
(D) Ionisation enthalpy of He will be same as Li^+

(1) T F T T

(2) F F T F

(3) F T T F

(4) T F F T

15) Give the correct order of initials true (T) or false (F) for given hypothetical species X^+, X and X^- :
(A) Z-effective decreases from X^+ to X^-
(B) Radii increases from X^- to X^+
(C) Ionisation enthalpy increases from X^- to X^+
(D) Electronegative decreases from X^+ to X^-

(1) T F T T

(2) T T F T

(3) F T T T

(4) T T T F

16) If an element X form the highest oxide of the formula XO_3, then it belongs to group :

(1) 13

(2) 14

(3) 15

(4) 16

17) Select the correct statement from the given and choose the correct option :

(A) Ga and Cs have very low melting point
(B) On going down in group-I, $\Delta_{eg}H$ becomes less negative
(C) s and p block elements are known as representative element except noble gases.
(D) The electronegative value of Be and Al on pauling scale is same

(1) A and B
(2) B and C
(3) C and D
(4) A, B, C and D

18) Match List-I with List-II :

	List-I (Elements)		**List-II (Property)**
(A)	Cl	(I)	Liquid non metal
(B)	Br	(II)	Maximum negative $\Delta_{eg}H$
(C)	H	(III)	Smallest element
(D)	Cs	(IV)	Electropositive element

Choose the correct answer from the options given below :
(1) A-II, B-I, C-III, D-IV
(2) A-III, B-II, C-IV, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-IV, B-I, C-III, D-II

19) Given below are two statements
Statement I : Be has lower IE than B.
Statement II : O has more IE than N and F.
Choose the correct answer from the option given below :

(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

20) An element X having configuration ns^2np^1 occurs in a short period of modern periodic table. The formula and nature of its oxide is :

(1) XO_3, amphoteric
(2) XO_3, acidic
(3) X_2O_3, amphoteric
(4) X_2O_3, basic

21) The difference between electron gain enthalpies will be maximum between :

(1) Ne and Cl
(2) Ar and Cl

(3) Ar and F

(4) Ne and F

22) Which of the following order is correct :

(1) $S^{2-} < Cl^{-} < K^{+}$: order of size

(2) K < Na < Mg < Ne : order of ionisation energy

(3) Mg < Al < Si : order of ionisation energy

(4) $Mg^{2+} < Al^{3+} < Na^{+}$: order of size

23) Match List-I with List-II.

	List-I (Ion/Compounds)		**List-II (Shape)**
(A)	H_3O^+	(I)	Distorted octahedral
(B)	NO_2	(II)	Trigonal pyramidal
(C)	XeF_6	(III)	Bent
(D)	$XeOF_4$	(IV)	Square pyramidal

Choose the correct answer from the options given below;

(1) A–III, B–II, C–IV, D–I

(2) A–II, B–III, C–I, D–IV

(3) A–III, B–II, C–I, D–IV

(4) A–II, B–III, C–IV, D–I

24) Which of the following pair of species with isostructural ?

(1) CO_2 and SO_2

(2) SiF_4 and SF_4

(3) XeF_2 and I_3^-

(4) SF_6 and XeF_6

25) Match the column :

	Column-I (Order)		**Column-II (Properties)**
(1)	B > Tl > Ga > Al > In	(a)	Electron affinity (EA)
(2)	Cl > F > Br > I	(b)	Electronagativity (EN)
(3)	F > Cl > Br > I	(c)	Ionisation energy (IP)
(4)	B < Ga < Al < In < Tl	(d)	Atomic radius

(1) 1 – a, 2 – b, 3 – d, 4 – c

(2) 1 – a, 2 – b, 3 – c, 4 – d

(3) 1 – c, 2 – a, 3 – b, 4 – d

(4) 1 – d, 2 – c, 3 – b, 4 – a

26) Read carefully the following statement-

(a) $Na_2SO_4 > Li_2SO_4$ (Solubility)
(b) $N_2 > B_2$ (no. of π bonds as per M.O.T.)
(c) $NH_3 > PH_3$ (Lewis basic strength)
(d) O–O < S–S (Bond energy)
(e) C – C > Si – Si (Bond energy)
How many statements are correct in the following.

(1) 2
(2) 3
(3) 4
(4) 5

27) For which of the following statements the van der Waals attractive force is the correct reason:

(1) I_2 is solid, Br_2 is liquid while F_2 and Cl_2 are gases.
(2) Acidic nature : HF < HCl < HBr < HI
(3) H_2O_2 is more viscous than H_2O.
(4) CH_3OH is more soluble in water than alcohols with higher molecular weight.

28) In allene (C_3H_4), the type(s) of hybridisation of the carbon atoms is (are) :

(1) sp and sp^3
(2) sp and sp^2
(3) only sp^2
(4) sp^2 and sp^3

29) The boiling point of p-nitrophenol is higher than that of o-nitrophenol because :

(1) p-nitrophenol has intermolecular hydrogen bonding while o-nitrophenol has intramolecular hydrogen bonding.
(2) p-nitrophenol has intramolecular hydrogen bonding while o-nitrophenol has intermolecular hydrogen bonding.
(3) $-NO_2$ group at p-position behaves in a different way than that of o-position.
(4) Hydrogen bonding exists in p-nitrophenol but no hydrogen bonding is present in o-nitrophenol

30) Identify the species having one π–bond and maximum number of canonical forms from the following:

(1) O_2
(2) SO_3
(3) CO_3^{-2}
(4) SO_2

31) Which of the following are isostructural pairs?
A. SO_4^{2-} and CrO_4^{2-}
B. $SiCl_4$ and $TiCl_4$

C. NH_3 and NO_3^-
D. BCl_3 and $BrCl_3$

(1) C and D only
(2) A and B only
(3) A and C only
(4) B and C only

32) Number of lone pairs of electrons in the central atom of SCl_2, O_3, ClF_3 and SF_6, respectively, are:

(1) 0, 1, 2 and 2
(2) 2, 1, 2 and 0
(3) 1, 2, 2 and 0
(4) 2, 1, 0 and 0

33) Given below are two statements :
Statement-I : SO_2 and H_2O both possess V–shaped structure.
Statement-II : The bond angle of SO_2 is less than that of H_2O.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is incorrect but Statement II is correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

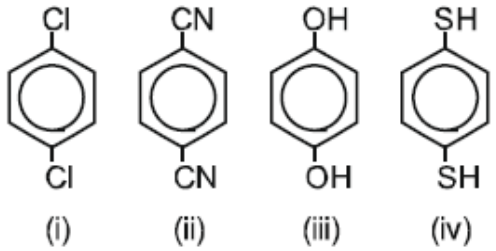
34) Choose the polar molecule from the following.

(1) CCl_4
(2) CO_2
(3) $CH_2 = CH_2$
(4) $CHCl_3$

35) Number of molecules/ions from the following in which the central atom is involved in sp^3 hybridization is :
$NO_3^-, BCl_3, ClO_2^-, BCl_4^-$

(1) 3
(2) 2
(3) 1
(4) 4

36) For which of the following molecule significant $\mu \neq 0$?



(1) Only (i)

(2) (i) and (ii)

(3) Only (iii)

(4) (iii) and (iv)

37) In which of the following pairs of molecules/ions both the species are not likely to exist?

(1) H_2^+, He_2^{2-}

(2) H_2^-, He_2^{2-}

(3) H_2^{+2}, He_2

(4) H_2^-, He_2^{2+}

38) Among the following molecules/ions,
$C_2^{2-}, N_2^{2-}, O_2^{2-}, O_2$
Which one is diamagnetic and has the shortest bond length?

(1) C_2^{2-}

(2) N_2^{2-}

(3) O_2

(4) O_2^{2-}

39) The bond order and the magnetic characteristics of CN^- are:

(1) 3, paramagnetic

(2) $2\frac{1}{2}$, paramagnetic

(3) 3, diamagnetic

(4) $2\frac{1}{2}$, diamagnetic

40) Maximum number of hydrogen bonds that can be formed by one H_2O molecule are :

(1) 2

(2) 4

(3) 6

(4) 1

41) Which of the following will have total number of d-electrons equal to the difference in the number of total p and s electrons?

(1) He

(2) Ne

(3) Ar

(4) Kr

42)

Match the column I with column II and mark the correct option :

Column I (Atomic number)		**Column II (Period, Group)**	
(A)	14	(i)	3, 14
(B)	53	(ii)	5, 2
(C)	38	(iii)	6, 10
(D)	78	(iv)	5, 17

(1) (A) → (ii), (B) → (iv), (C) → (iii), (D) → (i)

(2) (A) → (i), (B) → (iv), (C) → (ii), (D) → (iii)

(3) (A) → (iii), (B) → (ii), (C) → (i), (D) → (iv)

(4) (A) → (ii), (B) → (i), (C) → (iii), (D) → (iv)

43) Electronic configurations of few elements are given below. Mark the incorrect match.

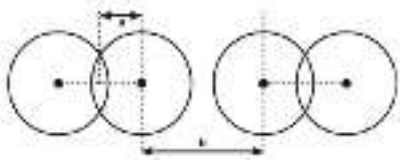
(1) $1s^2\ 2s^2\ 2p^5$ – Most electronegative element

(2) $1s^2\ 2s^2\ 2p^3$ – An element belonging to 3rd period and 15 group

(3) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^8\ 4s^2$ – d-block element

(4) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ – An element from 18th group

44) Comment on the radii shown as 'a' and 'b/2' in the figure of solid chlorine :



(1) a = Atomic radius, b/2 = Molecular radius

(2) a = Covalent radius, b/2 = van der Waals' radius

(3) a = Ionic radius, b/2 = Covalent radius

(4) a = Covalent radius, b/2 = Atomic radius

45) An element P has atomic number 56. What will be the formula of its halide?

(1) PX

(2) PX_2

(3) PX_3

(4) P_2X_3

BIOLOGY

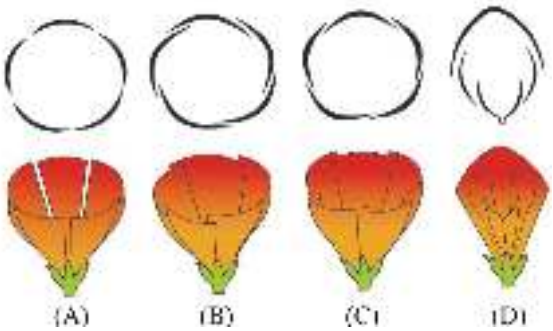
1) Consider the following four statements (a-d) and select the option which includes all the correct ones:-
(a) In *Rhizophora* pneumatophores are present which help to get oxygen for respiration.
(b) Prop roots are present in banyan tree for support.
(c) Stilt roots are present in maize and sugarcane. (d) Adventitious roots develop from radicle.

(1) Statement (a), (b) and (c)

(2) Statement (a), (b)

(3) Statement (c), (d)

(4) Statement (a), (c), (d)



2) Match the following aestivation with related plants :-
(i) *Cassia* (ii) Pea (iii) *Calotropis* (iv) Cotton

(1) A-iv, B-i, C-ii, D-iii

(2) A-iii, B-iv, C-ii, D-i

(3) A-iii, B-iv, C-i, D-ii

(4) A-iv, B-iii, C-i, D-ii

3) Which one of the following options gives the correct categorisation of six plants according to the type of placentation (a, b, c) they give out?

	(a) Axile	**(b) Basal**	**(c) Parietal**
(1)	Chinarose, lemon	Mustard, wheat	Sunflower, *Argemone*
(2)	Sunflower, lemon	China rose wheat	Mustard, *Argemone*
(3)	China rose, *Argemone*	Sunflower, wheat	Mustard lemon
(4)	China rose, lemon	Sunflower, wheat	Mustard, *Argemone*

(1) 1

(2) 2

(3) 3

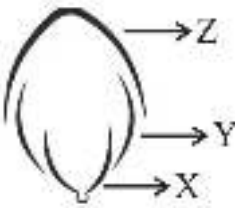
(4) 4

4) A sterile stamen is called :-

(1) Staminode
(2) Pistillode
(3) Synandrous
(4) Epipetalous

5)

Identify “X”, “Y” and “Z” in the given diagram ?



(1) X–Standard, Y–Wings, Z–Keel
(2) X–Wings, Y–Keel, Z–Standard
(3) X–Keel, Y–Wings, Z–Standard
(4) X–Keel, Y–Standard, Z–Wings

6) Trimerous flowers are commonly observed in :

(1) Solanaceae family
(2) Fabaceae family
(3) Brassicaceae family
(4) Liliaceae family

7) ⊕ and % indicate:

(1) Zygomorphic and actinomorphic flowers
(2) Actinomorphic and zygomorphic flowers
(3) Hypogynous and epigynous flowers
(4) None of these

8) Ovules are attached on central axis in a unilocular ovary, it is known as **A** placentation and it is present in **B**.
What are A & B respectively?

(1) A : Axile, B : Tomato
(2) A : Axile, B : *Primrose*
(3) A : Free central, B : Lemon
(4) A : Free central, B : *Dianthus*

9) **Assertion :-** Generally monocotyledonous seeds are endospermic
Reason :- Orchids have non endospermic seed

(1) Both assertion and reason are true the reason is a correct explanation of the assertion.

(2) Both assertion and reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but reason is false.

(4) Both assertions and reason are false.



10) Choose the correct floral formula with the help of given floral diagram :-

(1) $\oplus \, ♀ \, ♂ \, K_{2+2} C_4 A_{2+4} \underline{G}_{(2)}$

(2) $\% \, ♂ \, K_{(5)} C_{1+2+1+(2)} A_{(9)+1} \underline{G}_{1}$

(3) $\oplus \, ♀ \, ♂ \, K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G}_{(2)}$

(4) $Br \, \oplus \, ♀ \, ♂ \, \overset{\frown}{P_{(3+3)}} A_{3+3} G_{(3)}$

11)

Which one of the following options gives correct categorisation of six plants according to the type of flowers (X, Y, Z) they give out :

	(X) Hypogynous	**(Y) Perigynous**	**(Z) Epigynous**
(1)	Mustard, China rose	Rose, Peach	Guava, Cucumber
(2)	Mustard, Rose	China rose, Peach	Guava, Cucumber
(3)	Mustard, China rose	Rose, Guava	Peach, Cucumer
(4)	Mustard, Guava	Rose, Peach	China rose, Cucumber

(1) 1

(2) 2

(3) 3

(4) 4

12) **Statement-I :-** Variation in length of stamen filament within a flower is found in cruciferae.
Statement-II :- 5 sepals & 5 petals are found in all members of cruciferae.

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is not correct.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are incorrect.

13) Following posses alternate phyllotaxy except :-

(1) Mustard
(2) China rose
(3) Sunflower
(4) *Calotropis*

14) Identify the correct match from the columns I, II and III :

Column I	Column II	Column III
1. Zygomorphic flower	a. Radial symmetry	i. China rose
2. Actinomorphic flower	b. Irregular	ii. *Canna*
3. Asymmetric flower	c. Bilateral symmetry	iii. *Datura*
4. Hypogynous flower	d. Superior Ovary	iv. Gulmohur

(1) 2-a-iii, 1-c-iv, 3-b-i, 4-d-ii
(2) 1-c-iv, 2-d-iii, 3-b-i, 4-a-ii
(3) 1-d-iii, 2-b-iv, 3-c-i, 4-a-ii
(4) 2-a-iii, 1-c-iv, 3-b-ii, 4-d-i

15) Study the following statements carefully and give the answer :-
(A) Petioles allow leaf blades to flutter in wind.
(B) Veins provide rigidity to the leaf blade.
(C) A bud is present in the axil of both leaf and leaflet.
(D) In pinnately compound leaf, all leaflets are attached to a common point i.e. tip of petiole.

(1) Both A and C are correct
(2) Both A and D are correct
(3) Both C, D are wrong
(4) Both B and C are wrong

16) Rearrange the following zones as seen in the root in vertical section and choose the correct option for their sequence :-
A. Root hair zone
B. Zone of meristems
C. Rootcap zone
D. Zone of elongation

(1) C, B, D, A
(2) A, B, C, D
(3) D, A, C, B
(4) D, C, B, A

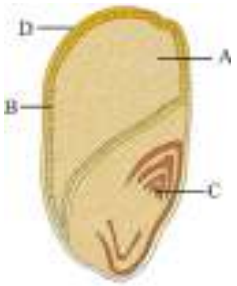
17) The phenomenon where in the ovary develops into a fruit without fertilisation is called _____.

(1) Parthenocarpy

(2) Apogamy
(3) Parthenogenesis
(4) Sexual reproduction

18) A few millimeters above the root cap is the region of :-

(1) Elongation
(2) Maturation
(3) Meristematic activity
(4) Root hairs



19) Identify the labels A, B, C and D in the figure given below :-

(1) A - Endosperm; B - Scutellum; C - Plumule; D - Seed coat
(2) A - Aleurone; B - Endosperm; C - Radicle; D - Coleorhiza
(3) A - Epithelium; B - Aleurone; C - Plumule; D - Seed coat
(4) A - Endosperm; B - Aleurone; C - Plumule; D - Seed coat and fruit wall

20) Which of the following is correct for the type of inflorescence given below ?



(1) The main axis terminates in a flower
(2) The main axis is limited in growth
(3) The flowers are borne in a basipetal order
(4) All are correct

21) G and G indicates :-

(1) Epigynous and hypogynous flowers
(2) Superior ovary and inferior ovary
(3) Presence and absence of gynoecium
(4) Fused and free gynoecium

22) Epipetalous condition is represented by :-

(1) $\overset{\frown}{C\ A}$

(2) $\overset{\frown}{P\ A}$

(3) $\overset{\frown}{A\ G}$

(4) $\overset{\frown}{G\ A}$



23) Identify the figure. This figure represent the :-

(1) Solanaceae

(2) Liliaceae

(3) Leguminosae

(4) Malvaceae

24) $\oplus\ \female\ K_{(5)}\ \overset{\frown}{C_{(5)}\ A_5}\ \underline{G}_{(2)}$

Which one of following options is correct in relation of above given floral formula ?

(1) Tomato, Chilli, *Petunia*

(2) *Asparagus*, Belladona, Potato

(3) *Aloe*, *Cochicum*, Tulip

(4) Mustard, Radish, *Capsella*

25) Which of these is an example for zygomorphic flower with imbricate aestivation?

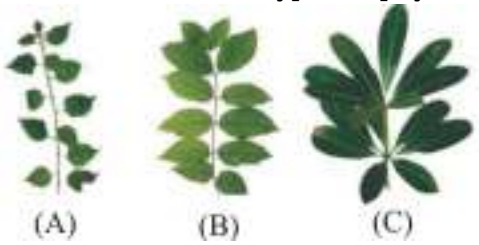
(1) *Canna*

(2) *Cassia*

(3) Cucumber

(4) *Calotropis*

26) The three sketches (A, B and C) given below represents three different types of phyllotaxy.



Select the option in which all are correctly identified :-

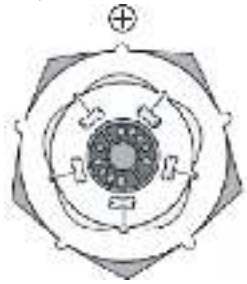
(1) A = Opposite, B = Whorled, C = Alternate

(2) A = Whorled, B = Opposite, C = Alternate

(3) A = Alternate, B = Opposite, C = Whorled

(4) A = Alternate, B = Whorled, C = Opposite

27) Choose the correct description of flower depicted in the floral diagram given below:



(1) United valvate sepal, united valvate petal, free epiphyllous stamen, unilocular ovary with parietal placentation.

(2) United valvate sepal, united valvate petals, free epiphyllous stamen, bilocular ovary with parietal placentation.

(3) United valvate sepal, united valvate petals, free epipetalous stamen, bilocular ovary with axile placentation

(4) United valvate sepal, united valvate petals, free epitepalous stamen, bilocular ovary with placentation free central

28) Placentation in which placenta develops at base of ovary and a single ovule is attached to it. It is known as ___A___ & present in ___B___ & ___C___ family. A, B & C are :

(1) A : Basal, B : Asteraceae, C : Fabaceae

(2) A : Basal, B : Asteraceae, C : Poaceae

(3) A : Free central, B : Asteraceae, C : Poaceae

(4) A : Marginal, B : Asteraceae, C : Fabaceae

29)

How many statements are correct for of family compositae?
(i) Most advanced type of inflorescence
(ii) Gynoecium has superior ovary
(iii) Disc florets are unisexual female flower

(iv) This is a type of racemose inflorescence

(1) (i) and (iv)

(2) (i) and (ii)

(3) (i), (ii) and (iii)

(4) only (i)

30) Match the followings :

A.	Actinomorphic	(i)	*Canna*
B.	Zygomorphic	(ii)	Tomato
C.	Asymmetrical	(iii)	Pea

(1) A-i, B-iii, C-ii

(2) A-ii, B-i, C-iii

(3) A-ii, B-iii, C-i

(4) A-i, B-ii, C-iii

31) Read the following statements :-
(A) Vascular bundles of monocot stem and leaves are surrounded by sclerenchymatous bundle sheath.
(B) Vascular cambium is involved in primary growth.
(C) Ring arrangement of vascular bundle is a characterstic feature of dicot stem.
(D) Intrafascicular cambium is present within a vascular bundle.
How many of above mentioned statement/s is/are true?

(1) Four

(2) Three

(3) One

(4) Two

32)

Find out incorrect statement about phloem :

(1) Phloem in angiosperm is composed of sieve tube elements, albuminous cell, phloem parenchyma and phloem fibres.

(2) A mature sieve element possesses a peripheral cytoplasm and a large vacuole but lacks a nucleus.

(3) In phloem parenchyma, plasmodesmatal connections exist between the cells.

(4) Phloem fibres of jute, flax and hemp are used commercially.

33) Read A to D and tell the correct order of components from outer side to inner side in a young dicot stem.
(A) Cortex (B) Pericycle
(C) Epidermis (D) Pith

(1) C, D, B, A

(2) A, B, C, D

(3) C, A, B, D

(4) C, B, A, D

34) The figure given below are different types of vascular bundles. Select the correct option according to diagram given below.



(1) A → Shows Concentric vascular bundle

(2) B → Shows radial vascular Bundle

(3) C → Shows Conjoint open vascular bundle

(4) A → Shows Conjoint closed vascular bundle.

35) Read the following four statements (A-D):
(A) Study of internal structure of plants is called anatomy.
(B) Different organs in a plant show differences in their internal structure.
(C) Growth in plants is largely restricted to specialised regions of active cell division called meristem.
(D) Primary meristem which occurs between mature tissues is known as intercalary meristem.
How many of the above statemens are correct ?

(1) Three

(2) Four

(3) One

(4) Two

36) Match the following :-

Column-I		Column-II	
a.	Axillary bud	i.	Occurs in layers below the epidermis in dicotyledonous plants
b.	Intercalary meristem	ii.	The meristems which occur at the tips of roots and shoots and produce primary tissues
c.	Apical meristem	iii.	They occur in grasses and regenerate parts removed by the grazing herbivores
d.	Collenchyma	iv.	Some cells left behind from shoot apical meristem

(1) a-i, b-ii, c-iii, d-iv

(2) a-ii, b-i, c-iv, d-iii

(3) a-iii, b-iv, c-i, d-ii

(4) a-iv, b-iii, c-ii, d-i

37) The functions of sieve tubes are controlled by nucleus of ?

(1) Specialised collenchymatous cells

(2) Companion cells

(3) Phloem parenchyma

(4) Sieve tube elements.

38) An anatomical structure consist of a large number of scattered vascular bundles and each bundle is surrounded by sclerenchymatous bundle sheath. Then the structure relates with ?

(1) Grass leaf

(2) Sunflower stem

(3) Grass stem

(4) Onion root

39) Match the column I and II :

Column I		Column II	
(A)	Bulliform cells	(i)	Dorsiventral leaf
(B)	Guard cells	(ii)	Aerating pore
(C)	Lenticels	(iii)	Stomata
(D)	Palisade parenchyma	(iv)	Isobilateral leaf

	A	B	C	D
(1)	(i)	(ii)	(iii)	(iv)
(2)	(iii)	(i)	(ii)	(iv)
(3)	(iv)	(i)	(ii)	(iii)
(4)	(iv)	(iii)	(ii)	(i)

(1) 1

(2) 2

(3) 3

(4) 4

40) Match the following :

(A)	Meristem	(i)	Sclereids
(B)	Parenchyma	(ii)	Actively dividing cells
(C)	Collenchyma	(iii)	Storage
(D)	Sclerenchyma	(iv)	Living mechanical tissue

	A	B	C	D
(1)	(ii)	(iii)	(iv)	(i)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(ii)	(iii)	(i)	(iv)
(4)	(iv)	(iii)	(ii)	(i)

(1) 1

(2) 2

(3) 3

(4) 4

41) **Assertion :** Apical meristem of root is subterminal
Reason : At the terminal end of root, root cap is present.

(1) Both Assertion and Reason true and reason is correct Explaination of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explaination of Assertion.

(3) Assertion is true but reason is false.

(4) Both Assertion and Reason are false.

42) (A) Water containing cavity is present in vascular bundle
(B) Vascular bundles are surrounded by sclerenchymatous bundle sheath.
(C) Phloem parenchyma is absent
(D) Sclerenchymatous hypodermis
These characters are associated with :-

(1) Maize stem

(2) Maize root

(3) Sunflower stem

(4) Maize leaf

43) Dicot root can be differentiated from monocot root in :

(1) Ring arrangement vascular bundle

(2) Polyarch xylem

(3) Do undergo secondary growth

(4) Both (1) and (3)

44) Consider the following statements and select the incorrect statement :-

(1) Axillary bud is derived from shoot apical meristem

(2) In sunflower stem, sclerenchymatous pericycle is located just above phloem bundle in semi lunar patches

(3) In conjoint vascular bundle phloem usually located on the inner side of xylem

(4) Phloem parenchyma is usually not present in monocot stem.

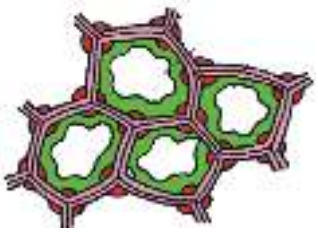
45) Epidermal cells modify to form :

(1) Trichomes only

(2) Trichomes, Root hairs only

(3) Trichomes, Root hair, Subsidiary cells

(4) Root hairs only



46) How many statements are incorrect for given diagrammatic sketch ? (i) Living mechanical tissue
(ii) Play significant role in pulp formation.
(iii) Wall becomes thick due to deposition of lignin and suberin
(iv) Present in petiole of leaf.

(1) (i), (iii) and (iv)

(2) (ii) and (iii)

(3) (ii) and (iv)

(4) (i), (ii) and (iii)

47) Which of the following tissue performs various functions like photosynthesis, storage, secretion etc.?

(1) Collenchyma and Sclerenchyma

(2) Parenchyma

(3) Xylem and phloem

(4) Sieve tube elements

48) The transverse section of plant material shows the following anatomical features under microscope: Palisade parenchyma towards adaxial surface and spongy parenchyma towards abaxial surface. Plant part material should be :-

(1) Isobilateral leaf

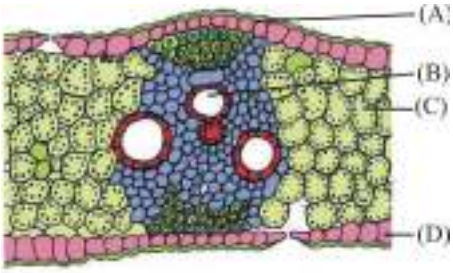
(2) Dorsiventral leaf

(3) Monocotyledonae stem

(4) Dicotyledonae stem

49)

Given below is the diagram of transverse section of leaf. Identify the parts labelled A, B, C and D and select the right option about them.



	(A)	(B)	(C)	(D)
(1)	Abaxial epidermis	Xylem	Spongy tissue	Adaxial epidermis
(2)	Adaxial epidermis	Xylem	Mesophyll	Abaxial epidermis
(3)	Adaxial epidermis	Phloem	Mesophyll	Abaxial epidermis
(4)	Abaxial epidermis	Xylem	Palisade tissue	Adaxial epidermis

(1) 1

(2) 2

(3) 3

(4) 4

50) In grasses, certain ____(a)____ cells modify themselves into large, empty, colourless cells. These cells are known as ____(b)____.

Select correct option for (a) and (b) ?

(1) a – Mesophyll; b – Guard cells
(2) a – Adaxial epidermal; b – Bulliform cells
(3) a – Abaxial epidermal; b – bulliform cells
(4) a – Meristem; b – Intercalary meristem

51) Which of the following statement about dicot stem is correct ?

(1) It has scattered vascular Bundles.
(2) Vascular bundles have sclrenchymatous bundle sheath.
(3) Vascular bundles are conjoint, collateral and closed.
(4) Hypodermis is collenchymatous.

52) Consider following features :-
(a) Hypodermis - Collenchymatous
(b) Vascular bundles conjoint open with endarch protoxylem
(c) Large number of pith cells
Select the correct option of plant organ which possess all the above characteristics.

(1) Monocot stem
(2) Dicot stem
(3) Monocot leaf
(4) Dicot leaf

53) Consider the following four (a-d) statements and select the option which includes all the incorrect statements only :-
(a) Companion cells help in maintaining the pressure gradient in sieve tubes.
(b) Phloem fibres are generally found in secondary phloem.
(c) Companion cells are present in gymnosperms.
(d) Companion cell is found in xylem.

(1) b, c, d
(2) a, c, d
(3) a, b, c, d
(4) c, d

54) Select erroneous (wrong) pair ?

(1) Parenchymatous conjunctive tissue – Dicotyledonous root
(2) Usually polyarch condition of xylem – Monocotyledonous root
(3) Usually diarch to tetrarch condition of xylem – Dicotyledonous root
(4) Scattered vascular bundles and each V.B. is surround by sclerenchymatous bundle sheath – Dictyledonous stem

55) Two statements are given below.
Statement-I : In dicotyledonous root, more than six xylem and phloem patches are present.

Statement-II : Monocot root do not undergo any secondary growth.
In the light of above statements choose the correct answer from the options given below.

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

56) "Protoxylem lies towards periphery and metaxylem lies towards the centre", such arrangement of primary xylem is called :

(1) Exarch, found in dicot stems

(2) Endarch, found in monocot stems

(3) Exarch, found in dicot roots

(4) Endarch, found in monocot roots

57) In dicot stem correct sequence of layers from outside to inner side is :-

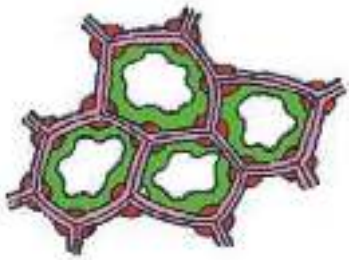
(1) Epidermis, pericycle, general cortex, endodermis, vascular bundle and pith

(2) Epidermis, pericycle, endodermis, general cortex, vascular bundle and pith

(3) Epidermis, general cortex, pericycle, endodermis, vascular bundle and pith

(4) Epidermis, general cortex, endodermis, pericycle, vascular bundle and pith

58) Which of the following is not a correct option regarding given figure :-



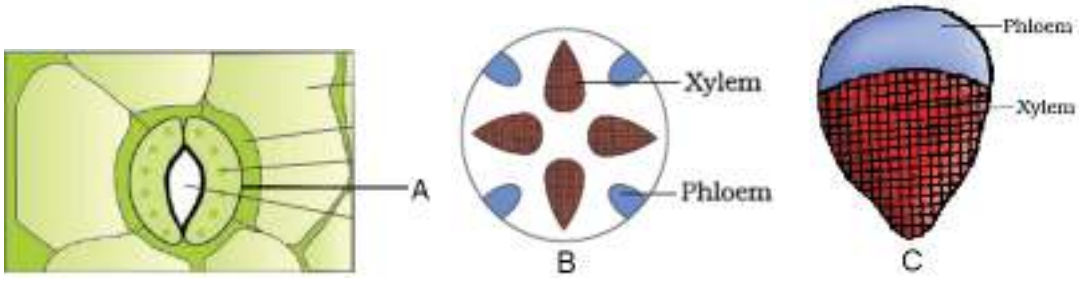
(1) It is found in stem of dicot below the epidermis

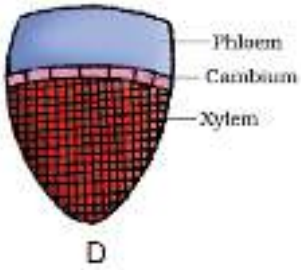
(2) It is commonly found in fruit walls of nuts, pulp of guava, pear & sapota, seed coat of legumes & leaves of tea.

(3) It is present in pedicel of a flower

(4) It is not present in tracheary element of xylem

59) Identify the (A), (B), (C) and (D) in the given figures and select the correct option :-





D

	(A)	(B)	(C)	(D)
(1)	Guard cells	Conjoint closed	Conjoint open	Radial condition
(2)	Subsidiary cells	Conjoint closed	Conjoint open	Conjoint condition
(3)	Guard cells	Radial condition	Conjoint closed	Conjoint Bicollateral
(4)	Guard cells	Radial condition	Conjoint closed	Conjoint open

(1) 1
(2) 2
(3) 3
(4) 4

60) Root hairs are the -

(1) Multicellular, elongations of epidermal cells
(2) Acellular, elongation of epidermal cells
(3) Unicellular, elongation of epidermal cells
(4) Multicellular, elongations of endodermal cells

61) The head of cockroach bears:

(1) One pair of compound eyes and two pairs of antennae
(2) One pair of compound eyes and a pair of antennae
(3) Three ocelli and no antennae
(4) No eyes but long mandibles only

62) Mouthparts of cockroach are of:

(1) Sucking type
(2) Biting and chewing type
(3) Piercing and sucking type
(4) Sponging type

63) In cockroach, the forewings are:

(1) Membranous and used for flight
(2) Leathery, opaque and protective (tegmina)
(3) Vestigial
(4) Absent in females

64) **Statement-I :** In cockroach, the total number of segments in abdomen is 10.
Statement-II : In females, the 7th sternum is boat shaped.

(1) Both statements are correct
(2) Both statements are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

65) A pair of short, thread-like ___**a**___ present in ___**b**___ cockroach.

(1) a - Anal styles, b - Male
(2) a - Anal styles, b - Female
(3) a - Anal cerci, b - Both male and female
(4) a - Anal styles, b - Both male and female

66) The gizzard in cockroach is part of:

(1) Midgut
(2) Foregut
(3) Hindgut
(4) Salivary glands

67) Malpighian tubules open at the junction of:

(1) Foregut and midgut
(2) Midgut and hindgut
(3) Only in rectum
(4) Only in crop

68) Number of pairs of spiracles in cockroach is:

(1) 8
(2) 10
(3) 12
(4) 6

69) In male cockroach, the accessory glands include:

(1) Only mushroom gland
(2) Mushroom gland and conglobate gland
(3) Only colleterial glands
(4) No accessory glands

70) In female cockroach, the _____ is a dark reddish to blackish brown capsule. How many _____ produce by female cockroach.

(1) Spermatophore, 1

(2) Ootheca, 14 – 16

(3) Ootheca, 9 – 10

(4) Spermatophore, 20

71) Development of *Periplaneta americana* is paurometabolous. What does this means ?

(1) Development occurs through a pupal stage.

(2) Development occurs through a nymphal stage.

(3) Development occurs through a larval stage.

(4) Development occurs through a only in water.

72) **Statement I:** Salivary glands of cockroach have reservoirs.
Statement II: Saliva contains amylase for starch digestion.

(1) Only Statement I is correct

(2) Only Statement II is correct

(3) Both statements are correct

(4) Both statements are incorrect

73) Which of the following statements are correct about the digestive system of cockroach?
1. Crop is a thin-walled storage sac.
2. Gizzard has internal chitinous teeth.
3. Hepatic caeca secrete cellulase only.
4. Rectum reabsorbs water.

(1) 1 and 2 only

(2) 1, 2 and 4 only

(3) 2 and 3 only

(4) All four

74) **Assertion (A):** Cockroach is uricotelic.
Reason (R): Malpighian tubules convert ammonia into uric acid.

(1) Both A and R are true and R is correct explanation of A

(2) Both A and R are true but R is not correct explanation of A

(3) A is true, R is false

(4) A is false, R is true

75) Match the Column

Column I (Structure)	Column II (Location/Number)
A. Spiracles	1. 10 pairs
B. Heart chambers	2. 13

C. Abdominal ganglia	3. 6
D. Hepatic caeca	4. 6–8

(1) A-1, B-2, C-3, D-4

(2) A-4, B-3, C-1, D-2

(3) A-1, B-3, C-4, D-2

(4) A-4, B-3, C-2, D-1

76) The part labelled “X” (a thick muscular structure with internal chitinous teeth at the end of foregut) is:



(1) Crop

(2) Gizzard

(3) Ileum

(4) Hepatic caeca

77) The body of an adult frog is divisible into

(1) Head, neck and trunk

(2) Head and trunk only

(3) Head, trunk and tail

(4) Cephalothorax and abdomen

78) Summer sleep in frog is called

(1) Hibernation

(2) Aestivation

(3) Diapause

(4) Torpor

79) Bile is secreted by which gland in frog?

(1) Pancreas

(2) Liver

(3) Salivary glands

(4) Gastric glands

80) Match the columns -

Column-I	Column-II
A. Tracheoles	i. Openings on cockroach body
B. Spiracles	ii. Site of gas exchange in cockroach
C. Cutaneous respiration	iii. Respiration via skin in frog
D. Pulmonary respiration	iv. Respiration via lungs in frog

(1) A-ii, B-i, C-iii, D-iv

(2) A-i, B-ii, C-iii, D-iv

(3) A-iv, B-iii, C-ii, D-i

(4) A-ii, B-iv, C-i, D-iii

81) Excretory products in -
(a) Cockroach
(b) Adult frog
(c) Tadpole

(1) a - Ammonia, b - Urea, c - Uric acid

(2) a - Uric acid, b - Urea, c - Ammonia

(3) a - Urea, b - Ammonia, c - Uric acid

(4) a - Uric acid, b- Ammonia, c - Urea

82) In male frog, sperms reach cloaca through

(1) Direct vas deferens

(2) Vasa efferentia → Bidder’s canal → ureter

(3) Oviduct

(4) Seminal vesicle only

83) **Statement-I :** Frogs have eyes that bulged out.
Statement-II : The eyes are covered by a nictitating membrane in water.

(1) Both statements are correct.

(2) Both statements are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

84) Buccopharyngeal respiration occurs

(1) Only in tadpole

(2) Through mouth cavity lining on land

(3) Only through lungs

(4) Only through skin

85) Pancreas in frog secretes

(1) Bile only

(2) Pancreatic juice containing digestive enzymes

(3) Mucus

(4) Hormones only

86) Which is correct statements :
I. Male and female frogs can be distinguished externally.
II. Frog is unisexual

(1) Only I correct

(2) Only II correct

(3) Both are correct

(4) Both are incorrect

87) Which of the following are correct about frog respiration?
1. Cutaneous respiration occurs on land and in water.
2. Pulmonary respiration uses lungs on land.
3. Gills are present in adult frog.
4. Buccopharyngeal respiration supplements on lands.

(1) 1 & 2 only

(2) 1, 2 & 4

(3) 1, 3 & 4

(4) All of these

88) **Assertion (A):** Frog heart has mixing of blood
Reason (R): Ventricle is single and undivided.

(1) Both A & R true, R explains A

(2) Both true, R does not explain A

(3) A true, R false

(4) A false, R true

89) Match Column I with Column II

Column I	**Column II**
A. Three-chambered heart	1. Male reproductive duct
B. Bidder’s canal	2. Common outlet
C. Nictitating membrane	3. Protection in water
D. Cloaca	4.Frog circulatory feature

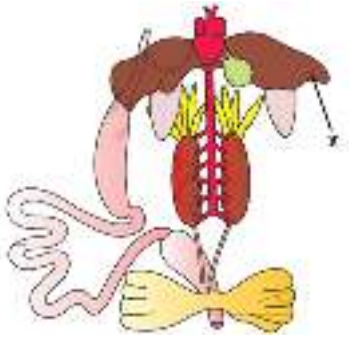
(1) A-1, B-2, C-3, D-4

(2) A-4, B-1, C-3, D-2

(3) A-1, B-3, C-4, D-2

(4) A-4, B-3, C-2, D-1

90) In the given diagram of frog internal organs, the structure labelled as 'X' -



(1) Secretes pancreatic juice

(2) Produces bile for fat digestion

(3) Stores urine

(4) Is part of nervous system

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	2	2	2	1	1	2	2	4	3	2	4	4	2	2	3	1	2	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	2	4	3	2	2	3	1	2	3	3	4	4	1	4	4	3	2	2	1
Q.	41	42	43	44	45															
A.	3	3	4	2	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	4	4	4	2	4	4	3	3	3	4	2	4	2	1	4	4	1	1	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	2	2	3	3	4	1	2	1	3	2	2	2	4	2	4	3	1	3	2
Q.	86	87	88	89	90															
A.	4	2	2	2	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	3	4	1	3	4	2	4	2	1	1	2	4	4	3	1	1	3	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	1	1	2	3	3	2	1	3	4	1	3	3	2	4	2	3	4	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	3	3	3	2	2	2	2	2	4	2	4	4	4	3	4	2	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	2	2	3	1	2	2	2	2	3	2	3	2	1	1	2	2	2	2	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	2	2	1	2	2	3	2	1	2	2										

SOLUTIONS

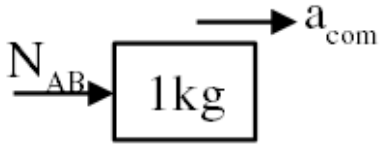
PHYSICS

1) **Question Explanation:**
The problem asks to find the case where the contact force between blocks A and B is maximum given that $m_A = m_B = 1$ kg.

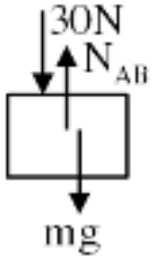
Concept:
This question is based on contact force, Newton's laws.

Solution:

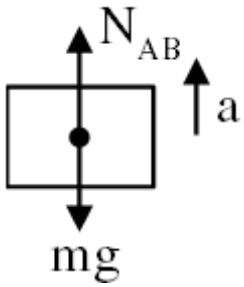
For (1) : Common acceleration $a_{com} = \frac{50}{1+1} = 25 m/s^2$



F_{BD} of B
$N_{AB} = m_A\ a_{com}$
$= 1 \times 25 = 25N$

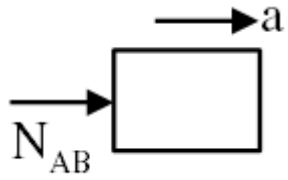


For (2) :
F_{BD} of A
$N_{AB} = 30 + 1 \times 10$
$= 40N$



For (3) :
F_{BD} of A
$N_{AB} - mg = ma$
$N_{AB} = 1 \times 25 + 1 \times 10$
NAB = 35 N

For (4) :



FBD of A
N_{AB} = ma
= 1 × 35
NAB = 35N

Final Answer:
The correct option is (2)

2) **Explaining :** Calculate the acceleration vector of a 4 kg mass when it is acted upon by two specific force vectors.
Concept : This question is based on the **Vector Form of Newton's Second Law**. The net force is the vector sum of all applied forces, and the acceleration is found by dividing this resultant vector by the mass
Solution :
Net force = $8\hat{i} + 4\hat{j} + 4\hat{k}$
$\vec{a} = \frac{\vec{F}}{m} = 2\hat{i} + \hat{j} + \hat{k}$
Final Answer : (3)

3) **Explaining :** To find distance travelled by block before it come to rest
Concept : This question is based on concept of stopping distance due to retarding force
Solution :
Force = mass × acceleration
u mg = ma
a = u g
a = 0.5 × 10 = **5 m/s²**
$v^2 = u^2 + 2as$
$(10)^2 = (0)^2 + 2\times5\times s$
100 = 10 × s
s = 10 m
Final Answer : (2)

4)

Explain the Question : Given that the linear momentum (p) of a particle varies with time (t) as
$p = a + bt + ct^2$ (where a, b, and c are constants), determine the correct statement regarding the particle's motion, specifically how the **Force** varies with time.
Concept : This question is based on Newton's Second Law of Motion - momentum.
Formula :
$F = \frac{dp}{dt}$
Solution : $P = a + bt + ct^2$

$F = \frac{dP}{dt} = b + 2ct$

$P = mV = a + bt + ct^2$

$v = \frac{a}{m} + \frac{b}{m}t + \frac{c}{m}t^2$

$v = \frac{ds}{dt} \Rightarrow \int ds = \int Vdt$

$S = \frac{a}{m}t + \frac{b}{m}\left(\frac{t^2}{2}\right) + \frac{c}{m}\left(\frac{t^3}{3}\right)$

Final answer : 2

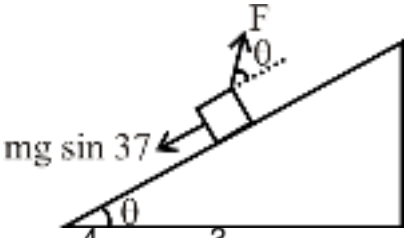
5) **Explaining :** To find average force on particle
Concept : This question is based on concept of newton's second law of motion.
Solution :
Change in momentum = m×dv = 2mv sin(θ/2)
Distance covered = Rθ
Time taken = Distance / Speed = (Rθ) / v
Average Force = Change in Momentum / Time Interval
F_{avg} = [mv^2 sin(θ/2)] / [(R θ/2]
Final Answer : (2)

6) **Explanation:**
Finding the acceleration of a 2 kg block on a smooth inclined plane (37°) pulled by a 20 N force at an angle of 37° to the incline.

Concept:

Newton's Second Law along an incline $\left(\sum F_{||} = ma\right)$

Solution:



$20 \times \frac{4}{5} - 20 \times \frac{3}{5} = 2a$

16 – 12 = 2a
4 = 2a
2 = a.

Final Answer: The correct option is (1).

7) **Explaining :** To determine the surface charge density on middle plate
Concept : This question is based on combination of capacitors
Solution :
From the given circuit we can see that there are two capacitors formed which are connected in parallel across the battery so total charge supplied by the battery to the middle plate is sum of

the charges on the two capacitors which is given as

$$q = (C_1 + C_2)V = \left[\frac{\varepsilon_1 \varepsilon_0 A}{d_1} + \frac{\varepsilon_2 \varepsilon_0 A}{d_2}\right] V$$

Surface charge density is given as

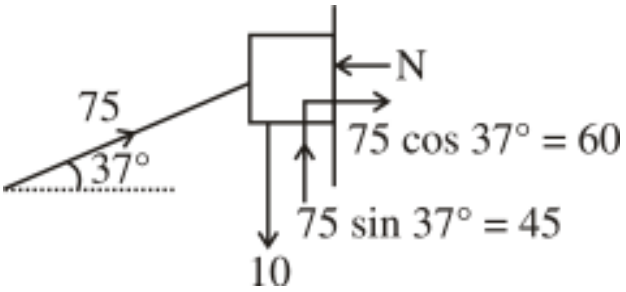
$$\sigma = \frac{q}{A} = \varepsilon_0 V \left[\frac{\varepsilon_1}{d_1} + \frac{\varepsilon_2}{d_2}\right]$$

Final Answer : (1)

8) **Explaining :** To calculate the magnitude of acceleration for a 1 kg block being pushed against a vertical wall by a 75 N force at a 37° angle.

Concept : This question is based on the concept of **Vertical Friction and Motion**. The acceleration is found by balancing vertical force components (push and gravity) against the frictional resistance generated by the horizontal normal force.

Solution :



N = 60

$f_{S_{max}} = f_k = 0.25 \times 60 = 15$

F in vertical direction

= F sin 37° – (mg+f_s)

45 - (10+15) = 20 N

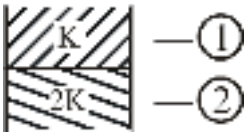
a = F/m = 20 m/s^2

Final Answer : (2)

9) **Explaining :** To choose the correct option when dielectrics are introduced between the plates of capacitor.

Concept : This question is based on parallel plate capacitor.

Solution :



$V_1 = V_2$

$E_1 d = E_2 d$

$E_1 = E_2$

$C_2 = 2C_1$

$$U_1 = \frac{1}{2} C_1 V_1^2 = \frac{1}{2} \text{Kcv}^2$$

$$U_2 = \frac{1}{2} C_2 V_2^2 = \frac{1}{2} (2\text{Kc})v^2$$

$U_2 > U_1$

Final Answer : (2)

10) **Explaining :** To choose the incorrect statement

Concept : This question is based on parallel plate capacitor
Solution :
The electric field between the plates of a capacitor is given as

$$E = \frac{2Q}{2A\varepsilon_0} + \frac{Q}{2A\varepsilon_0}$$

$$\Rightarrow E = \frac{3Q}{2A\varepsilon_0} = \frac{3}{2}\frac{Q}{Cd}$$

$$\Rightarrow Ed = \frac{3Q}{2C} = V$$

The force between the two plates is given as
F = QE

$$F = Q\left(\frac{2Q}{2A\varepsilon_0}\right)$$

$$F = \frac{Q^2}{A\varepsilon_0}$$

Energy stored between plate is given as

$$U = \frac{1}{2}\varepsilon_0 E^2 Ad$$

$$U = \frac{1}{2}\varepsilon_0\left(\frac{3Q}{2Cd}\right)^2 Ad$$

$$U = \frac{9}{8}\frac{Q^2}{C}$$

Thus option (4) is INCORRECT.
Final Answer : (4)

11) **Explaining :** Find the acceleration of a 20 kg trolley system being pulled by a 6 kg hanging block with kinetic friction.
Concept : This question is based on Newton's Laws and Kinetic Friction. The system's acceleration is determined by subtracting the frictional resistance from the weight of the hanging block and dividing by the total mass.
Solution :
$f_k = \mu N = 0.04 \times 20g = 8$ Newton

$$a = \frac{60-8}{26} = 2 m/s^2$$

Final Answer : (3)

12) **Explaining :** To determine, which change(s) will result in increase in capacitance.
Concept : This question is based on Parallel plate capacitor
Solution :
When x or z is replaced by a conductor, the effective separation between the capacitor plates decreases so its capacitance increases. If we replace y by a dielectric then it will act as another capacitor in series so overall capacitance will decrease. If x is replaced by a dielectric then also the capacitance of this section increases which will result in overall increase of capacitance. Thus option (A), (C) and (D) are correct.
Final Answer : (2)

13) **Explaining :** To find tension at a mid point on flexible chain
Concept : This question is based on concept of equilibrium
Solution :

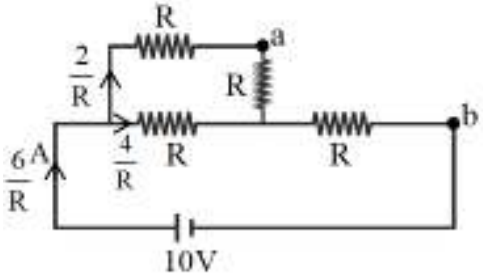
(1) **Vertical Balance:** At the support, the vertical component of tension (T sin θ) supports half the weight of the chain (W/2).
Equation: **T sin θ = W/2**
(2) Horizontal Tension: At the lowest point, the tension is purely horizontal and equal to the horizontal component of tension at the support (T cos θ).
Tension at midpoint (T_0) = T cos θ
(3) Substitution: Replace T from the first equation:
$T_0 = (W / 2 \sin \theta) * \cos \theta$
Since cos θ / sin θ = cot θ, the result is: **T_0 = (W/2) cot θ**
Final Answer : (4)

14) **Explaining :** To find dielectric constant of material
Concept : This question is based on parallel plate capacitor dielectric insertion.
Solution :
The charge on the capacitor before and after insertion of dielectric is given as
$Q_1 = C_1V$
$Q_2 = C_2V$

$$\Rightarrow \frac{50}{150} = \frac{1}{k}$$

$\Rightarrow k = 3$
Final Answer : (4)

15) **Explaining :** To find potential difference between plates of capacitor.
Concept : This question is based on RC circuit
Solution :



$$R_{eq} = \frac{2R.R}{3R} + R$$

$$= \frac{2R}{3} + R$$

$$= \frac{2R + 3R}{3} = \frac{5R}{3}$$

$$I = \frac{10}{5R} \times 3 = \frac{6}{R}$$

Potential difference between a and b is

$$V = \frac{2}{R} \times R + \frac{6}{R} \times R$$

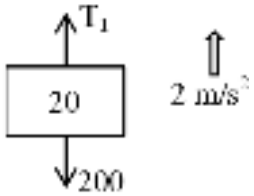
= 8 volt
Final Answer : (2)

16) **Explaining :** Determine the tension in the topmost rope while three hanging blocks (4kg, 6kg, and 10kg) accelerate upwards.
Concept : This question is based on Dynamics of Connected Hanging Bodies. The

tension in the primary support rope must balance the combined weight of all blocks and provide the necessary force for their shared upward acceleration.

Solution :

FBD of M_1 :



$T_1 - 200 = (4 + 6 + 10) \times 2$

$\Rightarrow T_1 = 240$

Final Answer : (2)

17) **Explaining :** Calculate the initial acceleration by identifying the net unbalanced force (restoring force of the spring) immediately after the supporting string is burned.

Concept : This question is based on the concept of **Laws of Motion and Spring Force**. While string tension disappears instantly when cut, the spring force remains unchanged at the initial moment, creating a net force that causes the block to accelerate immediately from its equilibrium position.

Solution :

vertical force F cos 53 = mg

kxcos 53 = mg

Horizontal force = F sin 53 = ma $a = \frac{kx \sin 53}{m} = g \tan 53 = \frac{4g}{3}$

Final Answer : (3)

18) **Explaining :** To find direction of action of friction force

Concept : This question is based on concept of direction of friction force is opposite to relative slipping

Solution :

Relative Velocity (v_{rel}) = $v_B - v_p$

$v_{rel} = (7\hat{i} - 7\hat{j}) - (3\hat{i} - 4\hat{j})$

$v_{rel} = (7 - 3)i + (-7+4)j$

v_{rel} = 4i - 3j

Friction will act in direction opposite to relative motion of block

Thus,

Unit Vector = (Opp. Vector) / (Magnitude)

Unit Vector = **(-4i + 3j) / 5**

Final Answer : (1)

19) **Explaining :** Apply Newton's Third Law to conclude that the bird and the airplane experience identical impact forces regardless of their mass or speed difference.

Concept : This question is based on the concept of **Newton's Third Law of Motion**. During any physical interaction or collision between two bodies, the action and reaction forces are always equal in magnitude and opposite in direction, acting on different objects.

Solution :

Due to Newton's third law.

Final Answer : (2)

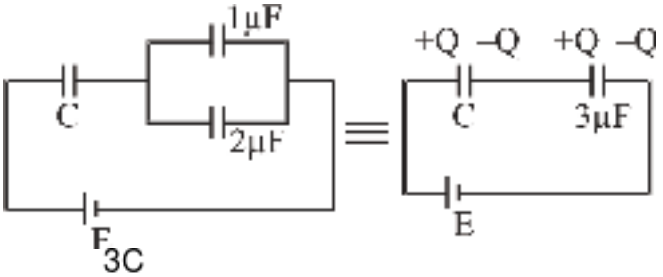
20) **Explaining :** To find tension in string connected by block on wedge
Concept : This question is based on concept of tension in string on wedge
Solution :
$F_{pull} = mg \sin(\theta)$
$F_{pull} = 12$ N
Normal Force (N) = $mg \cos(\theta) = 16$ N
$F_{friction}$ (max) = μ N = 12.8 N
Since the Maximum Friction (12.8 N) is greater than the Downward Force (12 N), the friction itself is enough to hold the block. Therefore, the string remains slack. Tension (T) = 0 N
Final Answer : (4)

21) **Explaining :** To find capacitance of each capacitor
Concept : This question is based on combination of capacitor
Solution :
The potential difference across each capacitor in series across a voltage V is equally divided so we have
$$V_C = \frac{V}{2}$$
Capacitance of capacitor is given as
$$C = \frac{Q}{V/2} = \frac{2Q}{V}$$
Thus option (3) is correct.
Final Answer : (3)

22) **Explaining :** To choose correct statements
Concept : This question is based on RC circuits
Solution :
When discharging starts then discharging current is given as
$$i = \frac{V}{R}e^{-\frac{t}{RC}}$$
At t = 0, discharging current is given as
$$i = \frac{V}{R}$$
Given that V and R are same for both capacitors, so the initial current in both capacitors is same but not zero.
During discharge, the instantaneous charge q at time t is given as
$q = q_0 e^{-t/RC}$
If $q = q_0/2$ at $t = t_1$,
$$\Rightarrow \frac{q_0}{2} = q^0 e^{-t/RC}$$
$\Rightarrow t_1 = RC \ln(2)$
If t_1 and t_1' be the times in which the two capacitors lose 50% of their charge, then we have
$$\frac{t_1}{t_1'} = \frac{RC_1 \ell n(2)}{RC_2 \ell n(2)} = \frac{C_1}{C_2} = \frac{1}{2}$$
$t_1 = t_1'/2$
This shows that C_1 loses 50% charge sooner than C_2 because it takes time t, which is half of t_1'.
Thus option (2) and (4) are correct.

Final Answer : (2)

23) **Explaining :** To choose the correct variation of Q_2 w.r.t. variation of C.
Concept : This question is based on charge on capacitor, Q = CV
Solution :



$C_{eq} = \frac{3C}{3+C}$ Q = Ceq.E

$Q_2 = \frac{2CE}{(C+3)}$

$Q_{2/\text{when C = 1μF}} = 2E\left[\frac{1}{4}\right] = \frac{E}{2}$

$Q_{2/\text{when C = 3μF}} = 2E\left[\frac{1}{2}\right] = E$

$\frac{dQ_2}{dC} = \frac{(C+3)(2E) - 2CE(1)}{(C+3)^2}$

$\frac{dQ_2}{dC} = \frac{6E}{(C+3)^2} > 0$

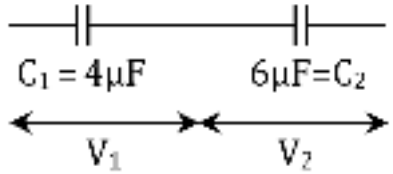
As $C \uparrow \frac{dQ_2}{dC} \downarrow$

Final Answer : (4)

24) **Explaining :** To find charge on 4μF capacitor in the network shown.
Concept : This question is based on voltage division rule.
Solution : $V_1 : V_2 = \frac{1}{4} : \frac{1}{6}$

$V_1 : V_2 = 6 : 4$

$V_1 = \frac{6}{6+4}(10) = 6V$



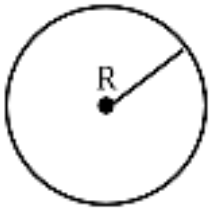
So, $Q_1 = C_1V_1 = 24\mu C$.
Final Answer : (3)

25) **Explaining :** To final the ratio of radii
Concept : This question is based on capacitance of spherical capacitor

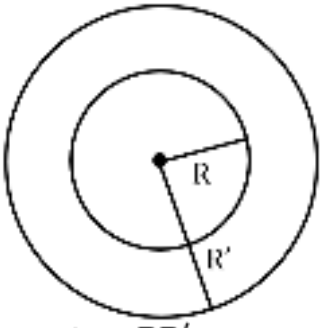
$C = 4\pi\varepsilon_0\frac{ab}{b-a}$

Solution :
for isolated sphere

$C = 4\pi\varepsilon_0 R$



for spherical capacitor with outer sphere earthed



$$C' = \frac{4\pi\varepsilon_0 RR'}{R'-R}$$
$$C' = nC$$
$$\frac{4\pi\varepsilon_0 RR'}{R'-R} = n(4\pi\varepsilon_0 R)$$
$$R' = nR' - nR$$
$$R' = \frac{nR}{n-1}$$
$$\frac{R'}{R} = \frac{n}{n-1}$$

Final Answer : (2)

26) **Explaining :** To find pseudo force on B by observer A

Concept : This question is based on concept of pseudo force

Solution :

Pseudo Force = -(Mass of Object B) × (Acceleration of Observer A)

$\mathbf{F_{pseudo} = - m_B\, a_A}$

$v_1 = 3(t^2)i - 4/3(t^6)j$

$a_A = dv_1/dt = (6t)i - (8t^5)j$

At **t = 1 sec**: a_A **= 6i - 8j**

Pseudo Force on B = -50 (6i - 8j)

$F_B = -100\,(3i - 4j)$

$F_B = 100\,(4j - 3i)$

Final Answer : (2)

27) **Explaining :** Find the acceleration of a 35 kg slab when a 100 N force is applied to a 7 kg block resting on its top.

Concept : This question is based on Two-Body Systems with Friction. When slipping occurs between the objects, the lower slab is accelerated specifically by the kinetic friction force transmitted through the contact surface.

Solution :

For a force of 100N on 7 kg block, relative motion will take place.

 The frictional force between 7 kg block and 35 kg block,

$f = \mu mg = 0.4 \times 10 \times 10N$

The acceleration of the slab of 35 kg is

$$a = \frac{0.4 \times 7 \times 10}{35} = 0.8 \text{ m/s}^2$$

 (3) is the correct answer

Final Answer : (3)

28) **Explaining :** To find equivalent capacitance

Concept : This question is based on combination of capacitors

Solution :

In the given circuit all the capacitors are in parallel, and so $C = 4 \times 8 = 32\ \mu F$

Final Answer : (1)

29) **Explaining :** To find the common potential

Concept : This question is based on combination of charged capacitors.

Solution :

Charge on capacitor A is given as

$q_1 = C_1 \times V$

$\Rightarrow q_1 = (15 \times 10^{-6})(100) = 15 \times 10^{-4}$ C

Charge on capacitor B is given as

$q_2 = C_2 \times V$

$\Rightarrow q_2 = (1 \times 10^{-6})(100) = 10^{-4}$ C

Capacitance of capacitor A after removing dielectric is given as

$C' = (C_1/k) = (15 \times 10^{-6}/15) = 1\ \mu F$

When both capacitors are connected in parallel their capacitance will be given as

$1\ \mu F + 1\ \mu F = 2\ \mu F$

Common potential of the combination is

$$V = \frac{q}{C}$$

$$\Rightarrow V = \frac{\left(15 \times 10^{-4}\right) + \left(1 \times 10^{-4}\right)}{2 \times 10^{-6}} = 800V$$

Final Answer : (2)

30) **Explaining :** To choose the correct statement regarding charge, potential difference and energy stored.

Concept : This question is based on Dielectric insertion between plates of capacitor.

Solution :

As the charging battery is disconnected, charge on plates of capacitor will remain constant and on inserting the dielectric slab capacitance increase so potential difference decreases and stored energy also decreases. Thus option (3) is correct.

Final Answer : (3)

31) **Explaining :** To find time in which y component of velocity will become zero.

Concept : This question is based on concept of newton's law of motion.

Solution :

$a_y = F_y / m = -5 / 5 = -1\ m/s^2$

$v = u + (a \times t)$

$0 = 40 + (-1 \times t)$

$0 = 40 - t$

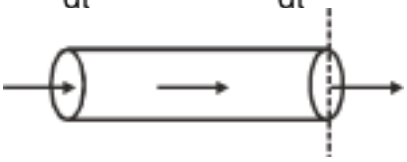
t = 40 s
Final Answer : (3)

32) **Explaining :** To find tension at a very specific point on rope.
Concept : This question is based on concept of tension due to continuous mass.
Solution :
M = 0.5 kg/m × 10 m
M = 5 kg
a = Applied Force / Total Mass
a = 25 N / 5 kg
a = 5 m/s^2
Tension = T = 1 kg × 5 m/s^2
T = 5 N
Final Answer : (4)

33) **Explaining :** Find the force at x = e by calculating acceleration using a = v * (dv/dx) and then multiply it by mass (F = m * a).
Concept : This question is based on Variable Acceleration. When velocity is given in terms of position (x), we use the differential form to find acceleration and then apply Newton Second Law.
Solution :
Velocity, v = ln(x)
a = v(dv/dx) thus a = ln(x) × (1/x)
a = ln(x) / x
For the Net Force to be zero, acceleration (a) must be 0: ln(x) / x = 0
x = 1 m
Final Answer : (4)

34) **Explanation :** To find force exerted on wall.
Concept : This question is based on of newton's second law of motion.
Solution :
Δp = 2 mv cosθ

$$F = \frac{dp}{dt} = (2v\cos\theta)\frac{dm}{dt}$$



The mass crossing an area in t would be
v × t × a × d = m ...(1)
For small element

$$\Rightarrow \frac{dm}{dt} = vad$$

$\Rightarrow$ F = $2av^2d\cos\theta$

$$F = 2 \times 2 \times 10^{-4} \times (3)^2 \times 1000 \times \frac{1}{2}$$

$$F = 18 \times 10^{-4} \times 10^3 N$$
$$F = 1.8N$$
Final Answer : (1)

35)

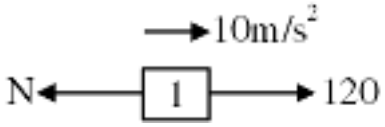
Question Explanation :
In the given arrangement we have to calculate the contact force between 1 kg and 4 kg.

Concept:
This question is based on Newton's 2nd law of motion

Solution:
Acceleration of the system

$$a = \frac{120-50}{7} = 10 m/s^2$$

Now for contact force between 1 kg and 4kg



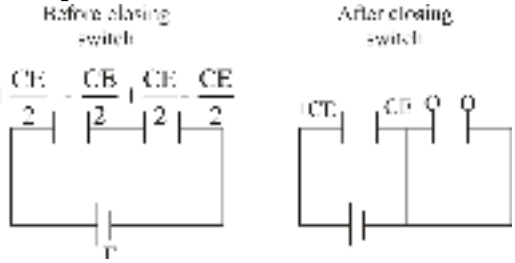
$\Rightarrow$ 120 – N = 1 × 10

$\Rightarrow$ $\boxed{N = 110\,Newton}$

Final Answer: option (4)

36) **Explaining :** To match the list- I with list-II according to the circuit shown.
Concept : This question is based on work done and heat loss.

Solution :



(P) Charge flown through battery after closing switch = (CE/2)

(Q) $(W.D)_{battery} = \left(\frac{CE}{2}\right) \times E = \left(\frac{CE^2}{2}\right)$

(R) Charge on capacitor A on closing switch = CE

(S) $(WD)_{Battery} = \Delta U + Heat$

$$\frac{CE^2}{2} = \left[\frac{1}{2}CE^2 - \frac{1}{2}\frac{CE^2}{4} \times 2\right] + Heat$$

$\Rightarrow$ Heat = $\frac{CE^2}{4}$

Final Answer : (4)

37) **Explaining :** To find maximum value of external source voltage
Concept : This question is based on breakdown of capacitors
Solution :
The maximum charge which can be supplied to the circuit is given by the two left capacitors which is given as
q = CV + 2 CV = 3 CV

The equivalent capacitance of system is

$$C' = \left[\frac{3C \times 6C}{3C + 6C}\right] = 2C$$

The maximum external voltage for this case can be given as

$$V = \frac{q}{C'} = \frac{3CV}{2C} = 1.5V$$

Final Answer : (3)

38) **Explaining :** To compare the area of plates of capacitor.

Concept : This question is based on $C = \frac{\varepsilon_0 A}{d}$.

Solution :

$$V \propto \frac{1}{C}$$

$V_2 > V_3 > V_1$

$C_2 < C_3 < C_1$

$A_2 < A_3 < A_1$

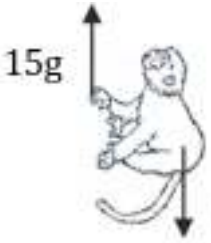
Final Answer : (2)

39) **Explaining :** Determine the minimum acceleration a 10 kg monkey must have while climbing a rope to lift a 15 kg mass off the ground.

Concept : This question is based on the concept of **Pulley and Tension Dynamics**. To lift a heavier load, the monkey must accelerate upwards to increase the rope's tension beyond the static weight of the block.

Solution :

To left the block off the ground T = 15 g (N = 0)



Applying NLM on monkey, we have 15 g – 10 g = 10 a

⇒ a = g/2

Final Answer : (2)

40) **Explaining :** To find the value of potential difference let a and b.

Concept : This question is based on Network analysis

Solution :

(1) Find the voltage across b and q (B_{bq})

In the bottom-most branch, C_1 and C_2 are in series.

Given V_{pq} = 2V across C_2.

Since Q = CV is constant for capacitors in series :$C_1V_{bp} = C_2V_{pq}$.

With $C_2 = 3C_1$, we get: $C_1V_{bp} = 3\ C_1(2V) \Rightarrow V_{bp} = 6V$

Total voltage between b and q is $V_{bq} = V_{bp} + V_{pq} = 6V + 2V = 8V$.

(2) Find the voltage between a and b (N)

The entire combination between b and q is in series with the capacitor C_1 (between a and b). We find the equivalent capacitance between b and q (C_{bq}):

* $C_{\text{series branch}} = \frac{C_1 \cdot 3C_1}{C_1 + 3C_1} = \frac{3}{4}C_1$

* $C_{bq} = C_2 + \frac{3}{4}C_1 = 3C_1 + 0.75C_1 = 3.75C_1$

Using the charge conservation ($Q_{ab} = Q_{bq}$)

$C_1 \cdot V_{ab} = C_{bq} \cdot V_{bq}$

$C_1 \cdot N = (3.75C_1) \cdot 8V$

$N = 3.75 \times 8 = \mathbf{30V}$

(3) Final Calculation

Substitute N = 30 into the required expression:

$\left(\frac{30 + 100}{100}\right) = \frac{130}{100} = \mathbf{1.3}$

Final Answer : (1)

41) **Explaining :** To find most suitable v-t curve for given condition.

Concept : This question is based on concept of slope variation of velocity time graph.

Solution :

If acceleration is constant, the velocity-time graph would be a straight line.

But here, **acceleration is increasing**. This means the slope of the velocity-time graph must increase over time.

This results in a **parabolic curve** (curving upwards) rather than a straight line.

The graph will show zero velocity for an initial period (until friction is overcome), followed by a curve that bends upwards because the increasing force creates increasing acceleration.

Final Answer : (3)

42) **Explaining :** To find the charge that flows through switch.

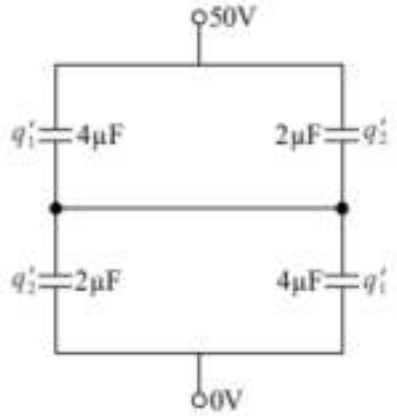
Concept : This question is based on switching circuits

Solution :

Initially on the left side of switch total charge is zero. After closing the switch circuit capacitance becomes 3 μF and final charge on the capacitors will be

$q_1' = 25(4) = 100\mu C$

$q_2' = 25(2) = 50\mu C$



The total charge on left side of switch is given as

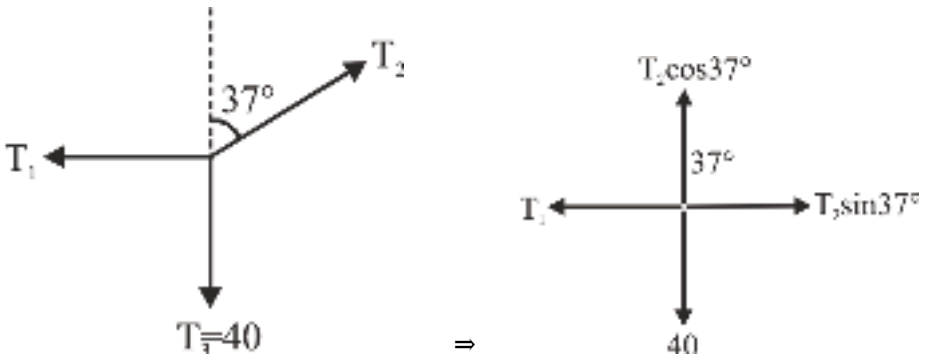
$q = q_1' + q_2'$

⇒ q = – 100 + 50 = – 50 μC

Thus the charge which flows from left to right through the switch is 50 μC..

Final Answer : (3)

43) **Explaining :** Determine the unknown tensions by resolving the forces at the junction into horizontal and vertical components and applying equilibrium conditions.
Concept : This question is based on the concept of **Translational Equilibrium**. For a mass-string system to remain in static balance, the vector sum of all tensions and weight at any point must be zero, which is solved by balancing components along the X and Y axes.
Solution :
$T_3 = 40$ N



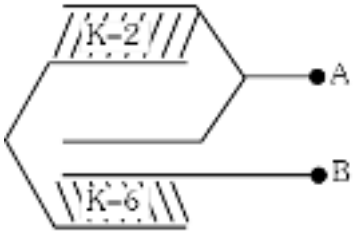
$T_2 \cos 37° = 40 \Rightarrow T_2 = 50$ N

$T_2 \sin 37° = T_1 \Rightarrow T_1 = T_2\left(\frac{3}{5}\right) = 30$ N
$\therefore T_1 = 30$ N, $T_2 = 50$ N, $T_3 = 40$ N
Final Answer : (4)

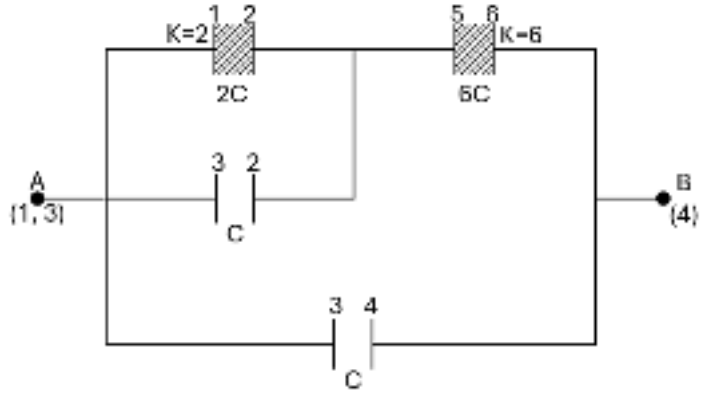
44)

Question Explanation : For the multiple plate setup (along with dielectric) , shown we need to find equivalent capacitance between A & B.

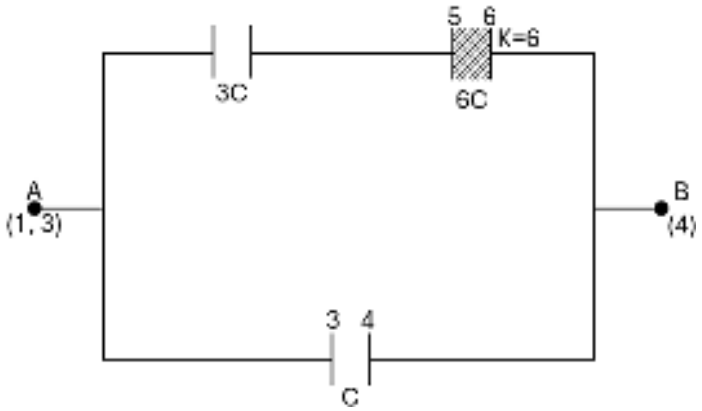
Concept : Combination of capacitors with dielectric.
Solution :



Circuit diagram :

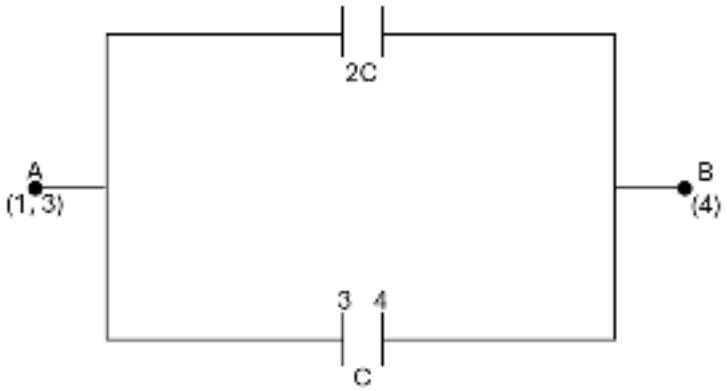


Capacitor 2C and C are in parallel combination:



3C and 6C are in series combination.

Resultant capacity = $\frac{3C \times 6C}{3C + 6C} = 2C$



2C and C are in series combination

C_{eq} = 2C + C = 3C

$C_{eq} = \frac{3A\varepsilon_0}{d}$

Final answer : Option (2)

45) **Explaining :** To find impulse imparted at t= 5 sec

Concept : This question is based on concept of calculation of impulse through change in momentum from graph

Solution :

Velocity just before t = 5s = 5 m/s

Velocity just after t = 5s = -7.5 m/s

Impulse = Change in Momentum

J = 1 × [(-7.5) - (5)] J = 1 × [-12.5] J = -12.5 N-s

Magnitude of Impulse: |J| = 12.5 N-s

Final Answer : (2)

CHEMISTRY

46) **Explaining :** Determine bond length order?

Concept : This question is based on bond length order

Solution :

Bond length order –

α. $CH_3 – CH_3$

β. $CH_2 = CH_2$

γ. $CH_2 = CH – CH = CH_2$ (conjugation)

correct order : α > γ > β
Final Answer : (1)

47) **Explaining :** Which of the following is correct order of acidic strength ?
Concept : This question is based on acidic strength
Solution :
Acidic strength ∝ electronegativity of central atom
Correct order of acidic strength –
(1) $H_2O_2 > H_2O$
(2) $H_3PO_3 < H_2SO_4$
(3) $H_2S < HBr$
Final Answer : (4)

48) **Explaining :** Which compound exist :
Concept : This question is based on existence and non existence
Solution :
IF_7 exists.
OF_4 (absence of vacant orbitals)
MnF_7, SCl_6 (steric crowding)
Final Answer : (4)

49) **Explaining :** Which of the following is incorrect order of boiling point.
Concept : This question is based on application of H-bonding
Solution :
Boiling point ∝ Molecular mass of molecules
Correct order of boiling point–
(1) $PH_3 < AsH_3 < NH_3 < SbH_3$
(2) $H_2S < H_2Se < H_2Te < H_2O$
(3) $HCl < HBr < HI < HF$
(4) $CH_4 < SiH_4 < GeH_4 < SnH_4$
Final Answer : (4)

50) **Explaining :** Find the correct statements about the polarity of molecules.
Concept : This question is based on polar and non polar molecules.
Solution :
(A) SF_4 (B) BF_3
(C) XeF_4 (D) SO_3
(A) is most polar molecule among the above given molecules because of presence of lone pair others have planar structure.
Final Answer : (2)

51) **Explaining :** The boiling point of water is exceptionally high because :
Concept : This question is based on effect of H-bonding on boiling point
Solution :
The boiling point of water is exceptionally high because water molecules associate due to hydrogen bonding.

Final Answer : (4)

52) **Explaining :** Evaluate assertion and reason
Concept : This question is based on VSEPR theory
Solution : Both Assertion and Reason are incorrect because in AB_3L_2 lone pairs are present at equatorial position.
Final Answer : (4)

53) **Explaining :** Which statements are true about noble gases :
Concept : This question is based on atomic radius
Solution :
Vander Waals radii > Covalent radii > Metallic radius
Noble gas are stable gases so they exist as monoatomic gases.
Final Answer : (3)

54) **Explaining :** In the given graph of ionisation enthalpy. Find out the element x, y and z respectively :
Concept : This question is based on ionization energy
Solution :
Across the period on moving L → R
I.E. increasing and down the group IE decreases.
Final Answer : (3)

55) **Explaining :** Greatest amount of energy involves in the transformation of :
Concept : This question is based on ionization energy
Solution :
Greatest amount of energy involves in the transformation of $Li^+ \rightarrow Li^{+2} + e^-$
$Li^+ (1s^2)$ - removing e^- from $1s^2$ requires greatest amount of energy
Final Answer : (3)

56) **Explaining :** Non metals are usually solids or gases at room temperature with low melting point and boiling point, the exceptions are :
Concept : This question is based on physical properties of elements
Solution :
Non metals are usually solids or gases at room temperature with low melting point and boiling point with the exception of Carbon, Boron.
Br_2 is liquid at room temperature. Carbon and Boron exist as 3D covalent solid & they have exceptionally high melting point.
Final Answer : (4)

57) **Explaining :** The order in which the energy of the available orbitals in 5th period are :
Concept : This question is based on filling of electrons in available orbitals in n^{th} period
Solution :
The order in which the energy of the available orbitals in 5th period are 5s < 4d < 5p
Final Answer : (2)

58) **Explaining :** How many elements of 4th period will have one or more electrons in their 4d subshell :
Concept : This question is based on electronic configuration
Solution : In 4th period electron goes in penultimate d sub shell.
Final Answer : (4)

59) **Explaining :** Select the statement from the given and choose the initials as (T) for true and (F) for false :
Concept : This question is based on multiple concept
Solution :
In p-block, metals, metalloids and non metals are present.
ΔE_{eg1} of N, Be, noble gases, Mg is positive.
IE : $Li^+ > He$ (Isoelectronic species)
Final Answer : (2)

60) **Explaining :** Give the correct order of initials true (T) or false (F) for given hypothetical species X^+, X and X^- :
Concept : This question is based on application of effective nuclear charge.
Solution :

$$Z_{eff} \propto \frac{+ve}{-ve} \propto \frac{1}{radius} \propto IE \propto EN$$

- Z-effective decreases from X^+ to X^-
- Radii decreases from X^- to X^+
- Ionisation enthalpy increases from X^- to X^+
- Electronegative decreases from X^+ to X^-

Final Answer : (1)

61) **Explaining :** If an element X form the highest oxide of the formula XO_3, then it belongs to group :
Concept : This question is based on maximum valency corresponding to the group number in covalent oxides.
Solution :
Group 16 elements form two type of oxide.
EO_3, EO_2
Final Answer : (4)

62) **Explaining :** Select the correct statement from the given and choose the correct option :
Concept : This question is based on properties of elements.
Solution :
- Ga and Cs have very low melting point
- On going down in group-I, $\Delta_{eg}H$ becomes less negative
- s and p block elements are known as representative element except noble gases.
- The electronegative value of Be(1.5) and Al(1.5) on pauling scale is same

Final Answer : (4)

63) **Explaining :** Match List-I with List-II :
Concept : This question is based on physical property

Solution :

	List-I (Elements)		**List-II (Property)**
(A)	Cl	(I)	Maximum negative $\Delta_{eg}H$
(B)	Br	(II)	Liquid non metal
(C)	H	(III)	Smallest element
(D)	Cs	(IV)	Electropositive element

Final Answer : (1)

64) **Explaining :** Analyse both the statments
Concept : This question is based on ionization energy
Solution :
• Be has higher IE than B.
• O has lesser IE than N and F.
Final Answer : (1)

65) **Explaining :** An element X having configuration ns^2np^1 occurs in a short period of modern periodic table. The formula and nature of its oxide is :
Concept : This question is based on nature of oxides of G-13 elements
Solution :
• Period 2nd and 3rd having 8 elements and known as short period.
• ns^2np^1 will be element of boron family.
B_2O_3 - Acidic , Al_2O_3 - Amphoteric
Final Answer : (3)

66) **Explaining :** The difference between electron gain enthalpies will be maximum between :
Concept : This question is based on electron gain enthalpy
Solution : The $\Delta_{eg}H$ for Ne is maximum positive and for Cl is maximum negative.
Final Answer : (1)

67) **Explaining :** Which of the following order is correct :
Concept : This question is based on ionic size and ionization energy
Solution :
$S^{2-} > Cl^- > K^+$: order of size
K < Na < Mg < Ne : order of ionisation energy
Mg > Al < Si : order of ionisation energy
$Mg^{2+} > Al^{3+} < Na^+$: order of size
Final Answer : (2)

68) **Explaining :** Match List-I with List-II.
Concept : This question is based on shape of molecules.
Solution :

	(Ion/Compounds)		**(Shape)**

(A)	H_3O^+	(II)	Trigonal pyramidal
(B)	NO_2	(III)	Bent
(C)	XeF_6	(I)	Distorted octahedral
(D)	$XeOF_4$	(IV)	Square pyramidal

Final Answer : (2)

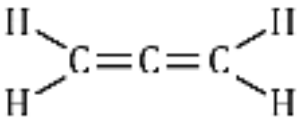
69) **Explaining :** Which of the following pair of species with isostructural ?
Concept : This question is based on isostructural
Solution :
XeF_2 and I_3^- pair of compounds with isostructural. Both are linear in structure.
Final Answer : (3)

70) **Explaining :** Match the given elemental orders with their corresponding chemical properties.
Concept : This question is based on periodic trends.
Solution:
(a) B > Tl > Ga > Al > In → Ionisation energy of (13th group)
(b) Cl > F > Br > I → Electron affinity (17th group)
(c) F > Cl > Br > I → Electronegativity (17th group)
(d) B < Ga < Al < In < Tl → Atomic radius
Final Answer: (3)

71) **Explaining :** How many statements are correct in the following.
Concept : This question is based on Multiple concept
Solution : All the statements are correct.
Final Answer : (4)

72) **Explaining :** For which of the following statements is the van der Waals attractive force the correct reason:
Concept : This question is based on weak forces
Solution :
I_2 is solid, Br_2 is liquid while F_2 and Cl_2 are gases.
This statement is correct, as the variation in physical states of halogens is directly attributed to the increases in Vander waal's attractive forces with increasing molecular mass, size and polarizability.
Final Answer : (1)

73) **Explaining :** In allene (C_3H_4), the type(s) of hybridisation of the carbon atoms is (are) :
Concept : This question is based on hybridisation.
Solution :
sp and sp^2

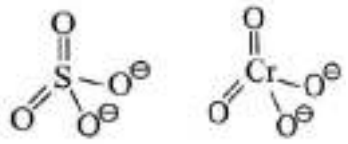


Final Answer : (2)

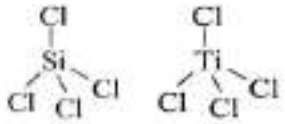
74) **Explaining :** The boiling point of p-nitrophenol is higher than that of o-nitrophenol because :
Concept : This question is based on application of H-bonding
Solution :
The boiling point of p-nitrophenol is higher than that of o-nitrophenol because p-nitrophenol has intermolecular hydrogen bonding while o-nitrophenol has intramolecular hydrogen bonding.
Final Answer : (1)

75) **Explaining :** Identify the species having one π–bond and maximum number of canonical forms from the following:
Concept : This question is based on resonance
Solution :
Among all, only O_2 and CO_3^{-2} have one π–bond but CO_3^{-2} has more resonating structures i.e., three
Final Answer : (3)

76) **Explaining :** Which of the following are isostructural pairs?
Concept : This question is based on isostructural pairs
Solution :
A. SO_4^{2-} and CrO_4^{2-}



B. $SiCl_4$ and $TiCl_4$



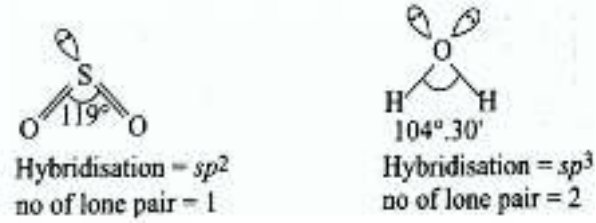
Final Answer : (2)

77) **Explaining :** Number of lone pairs of electrons in the central atom of SCl_2, O_3, ClF_3 and SF_6, respectively, are:
Concept : This question is based on VSEPR Theory.
Solution : -

$$H = \frac{1}{2}[V + X - C + A]$$

2, 1, 2 and 0
Final Answer : (2)

78) **Explaining :** Analyze both the statements
Concept : This question is based on VSEPR theory
Solution :



Bond angle of SO_2 is greater than that of H_2O due to higher repulsion of multiple bonds..

Final Answer : (2)

79) **Explaining :** Choose the polar molecule from the following.

Concept : This question is based on polarity of a molecule.

Solution :

In CCl_4, CO_2 and $CH_2{=}CH_2$, individual bond moments cancel out to give a net zero dipole moment. In $CHCl_3$, because it is not a symmetrical molecule, vector sum of bond moments is not zero i.e., it is as polar molecule.

Final Answer : (4)

80) **Explaining :** Number of molecules/ions from the following in which the central atom is involved in sp^3 hybridization is :

Concept : This question is based on hybridization.

Solution :

NO_3^- - sp^2 $\quad$ ClO_2^- - sp^3

BCl_3 - sp^2 $\quad$ BCl_4^- - sp^3

Final Answer : (2)

81) **Explaining :** For which of the following molecule significant $\mu \neq 0$?

Concept : This question is based on dipole moment.

Solution : $\mu \neq 0$ for (iii) and (iv) because of presence of lone pairs on oxygen & sulphur.

Final Answer : (4)

82) **Explaining :** In which of the following pairs of molecules/ions both the species are not likely to exist?

Concept : This question is based on existance & non-existance of molecules.

Solution :

$$BO = \frac{N_b - N_a}{2}$$

Final Answer : (3)

83) **Explaining :** Which one is diamagnetic and has the shortest bond length?

Concept : This question is based on application of bond length.

Solution : -

$$BO = \frac{N_b - N_a}{2}$$

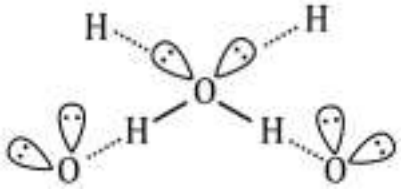
$$B.O. \propto \frac{1}{\text{bond length}}$$

Thus, C_2^{2-} has shortest bond length and diamagnetic.

Final Answer : (1)

84) **Explaining :** The bond order and the magnetic characteristics of CN^- are:
Concept : This question is based on bond order and magnetic behaviour of ions or molecules.
Solution : -
CN^- is a 14 electron system.
B.O. is $\frac{1}{2}[10-4]=3$, no unpaired electron.
Final Answer : (3)

85) **Explaining :** Maximum number of hydrogen bonds that can be formed by one H_2O molecule are :
Concept : This question is based on H-bonding
Solution :



Final Answer : (2)

86) **Explaining :** Which of the following will have total number of d-electrons equal to the difference in the number of total p and s electrons?
Concept : This question is based on electronic configuration.
Solution : -
The first inert gas which contains d electrons is in 4th
period i.e., Kr with atomic no. = 36
E.C. of Kr = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6$
Total number of d-electrons = 10
Total number of p-electrons = 6 + 6 + 6 = 18
Total number of s-electrons = 2 + 2 + 2 + 2 = 8
Difference in p and s electron is = 18 – 8 = 10
Final Answer : (4)

87) **Explaining :** Match the column I with column II and mark the correct option :
Concept : This question is based on atomic number.
Solution :
(A) 14 → (i) 3, 14
(B) 53 → (iv) 5, 17
(C) 38→ (ii) 5, 2
(D) 78 → (iii) 6, 10
Final Answer : (2)

88) **Explaining :** Electronic configurations of few elements are given below. Mark the incorrect match.
Concept : This question is based on periodic properties.
Solution : $1s^2\ 2s^2\ 2p^3$ – An element belonging to 2nd period and 15 group
Final Answer : (2)

89) **Explaining :** Comment on the radii shown as 'a' and 'b/2' in the figure of solid chlorine :
Concept : This question is based on atomic radius.
Solution :

$$\text{Covalent radius} = \frac{\text{internuclear distance between two covalently bonded atoms}}{2}$$

$$\text{Vander waal radius} = \frac{\text{internuclear distance between two nuclei of adjacently placed atoms in solid state of a noble gas or molecule}}{2}$$

Final Answer : (2)

90) **Explaining :** An element P has atomic number 56. What will be the formula of its halide?
Concept : This question is based on ionic compound formation.
Solution : The element will be of group IInd so formula of halide will be PX_2.
Final Answer : (2)

BIOLOGY

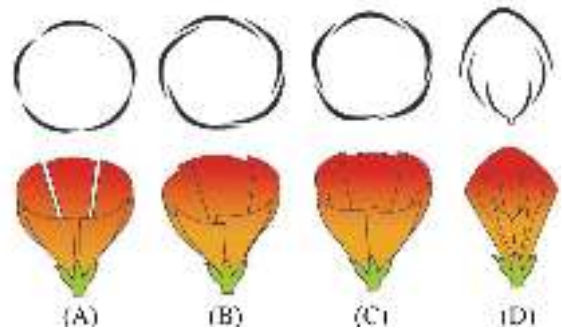
91) **Solution :** In marshy/swampy areas, there is scarcity of oxygen so. Some branches of tap roots of the plants which grow in this region, grow vertically upward and come to the surface. These roots are called pneumatophores which have minute pores by which air enters and plant gets oxygen for respiration, Pneumatophores are negatively geotropic. e.g. *Rhizophora* (Mangrove plant).

Prop roots or pillar roots : These hanging roots arise from branches of plant and grow down wards towards the soil. These roots support the tree. e.g. Banyan (*Ficus benghalensis*)

Stilt roots or brace roots : These roots arise from lower node and enter in the soil. These roots are supporting roots. e.g. Maize, sugarcane.
Adventitious Roots : In some plants, like grass, *Monstera* and the Banyan tree, roots develop from parts of the plant **other than the radicle** and are known as adventitious roots.

Final Answer : (1)

92) **Question Explanation :** Match the following aestivation types with the correct plants.



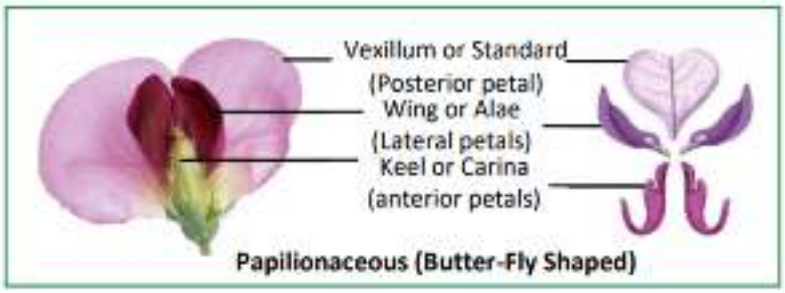
(i) *Cassia* (ii) Pea (iii) *Calotropis* (iv) Cotton
Concept : This question is based on Aestivation.
Solution :
i (*Cassia*) → Imbricate → (C)
ii (Pea) → Vexillary → (D)
iii (*Calotropis*) → Valvate → (A)

iv (Cotton) → Twisted → (B)
Final Answer : (3)

93) **Solution :**
Axile : In this type of placentation ovary is multilocular and the ovules are borne on the central axis. Number of chambers are equal to the number of carpels fused. e.g. China rose, Lemon, Tomato.
Basal : In this type of placentation a single ovule is attached at the base of ovary. e.g. Marigold, Sunflower (Asteraceae/ Composite family), Gramineae / Poaceae family.
Parietal : In this type of placentation the ovules develop on the inner wall of the ovary or on the peripheral part. e.g. Mustard and *Argemone*.
Final Answer : (4)

94) **Solution :** Staminode : Sterile stamen is called staminode. e.g. *Salvia*, Caesalpinoidae (*Cassia*, gulmohur, tamarind)
Final Answer : (1)

95) **Solution :** X–Keel, Y–Wings, Z–Standard



Final Answer : (3)

96) **Solution :** A flower may be trimerous, tetramerous or pentamerous when the floral appendages are in multiple of 3,4 or 5 respectively. In dicots flowers are usually pentamerous while in monocots flowers are trimerous. Like Liliaceae.
Final Answer : (4)

97) **Solution :**

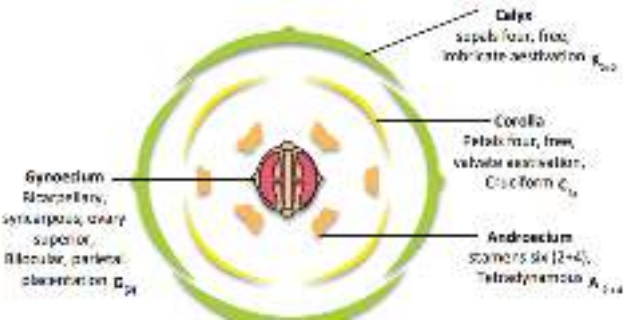
Actinomorphic	=	⊕
Zygomorphic	=	% or ⏀ or ÷

Final Answer : (2)

98) **Solution : Free Central/Central** : In this type of placentation the ovary is unilocular (septa are absent) and the ovules are borne on the central axis. e.g. *Primrose*, *Dianthus*
Final Answer : (4)

99) **Solution :** Generally monocotyledonous seeds are endospermic, orchids are also monocot but during the embryo development complete endosperm consumed. So we can say that the orchids is a exception & have non endospermic seed.
Final Answer : (2)

100) **Solution :**



Final Answer : (1)

101) **Solution :**
(A) Hypogynous Flower : When gynoecium occupies the highest position while the other parts like petals, sepals and stamens are situated below the ovary, then the flower is called hypogynous and in this condition ovary will be superior. e.g. Mustard, China rose, Brinjal, Mango.
(B) Perigynous Flower : In it thalamus grows upwardly and form a cup shaped structure. On the margin or rim of thalamus floral parts are attached except gynoecium, which lies at the basal part or in the centre. So, in this condition gynoecium is situated below the other floral parts. But ovary in this condition is said to be half inferior. e.g. Rose, Plum, Peach.
(C) Epigynous Flower : When the margin of thalamus grows upward enclosing the ovary completely and getting fused with it and other parts of flower like petals, sepals & stamens are situated above the ovary, then the ovary is said to be inferior and rest of the floral parts superior. e.g. Guava, Apple, Cucumber and the ray florets and disc florets of sunflower.

Final Answer : (1)

102) **Solution :** There may be a variation in the length of filaments within a flower, as in *Salvia* and mustard (Cruciferae family).
Calyx - Sepals 4, polysepalous, sepals are arranged in two whorls 2 outer and 2 inner, imbricate aestivation.
Corolla - Petals 4, polypetalous, valvate aestivation and cruciform/cross form.
Final Answer : (2)

103) **Concept :** This question is based on Leaf-phyllotaxy
Solution : In *Calotropis* opposite type of phyllotaxy are found.
Final Answer : (4)

104) **Solution:** 2-a-iii, 1-c-iv, 3-b-ii, 4-d-i

(A) Actinomorphic/radial/regular : When a flower can be divided into two equal radial halves by any vertical plane or radial plane passing through the centre, then it is said to be actinomorphic flower e.g. Mustard, *Datura*, Chilli.

(B) Zygomorphic/bilateral : When a flower can be divided into two equal (similar) halves only by one particular vertical plane, then it is said to be zygomorphic flower. e.g. Pea, Bean, Gulmohur, *Cassia*.

(C) Asymmetrical/irregular : When a flower cannot be divided into two equal (similar) halves from any vertical plane passing through the centre, then it is said to be asymmetrical flower. e.g. *Canna*.

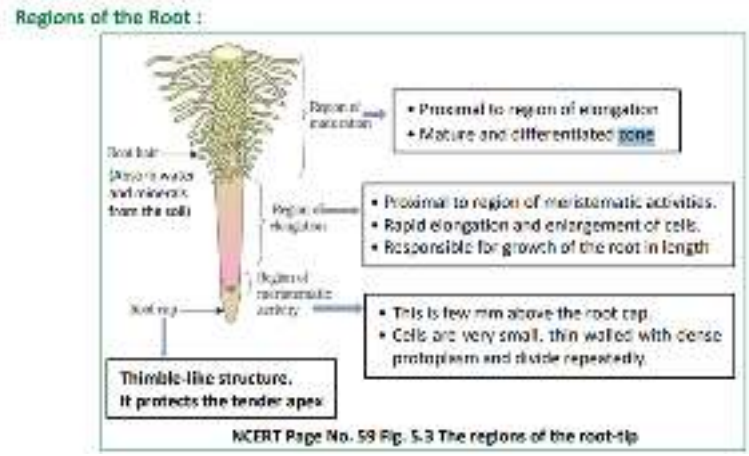
Final Answer: (4)

105) **Solution :**

A. Long thin flexible petioles allow leaf blades to flutter in wind, thereby cooling the leaf and bringing fresh air to leaf surface.
B. Veins provide rigidity to the leaf blade and act as channels of transport for water, minerals and food materials.
C. A bud is present in the axil of petiole in both simple and compound leaves, but **not** in the axil of leaflets of the compound leaf.
D. **Pinnately Compound Leaf** : In this type of leaf, leaf blade (lamina) is incised upto the mid rib and mid rib is known as rachis. A number of leaflets are present on a common axis, the rachis. e.g. Neem

Final Answer : (3)

106) **Solution :**

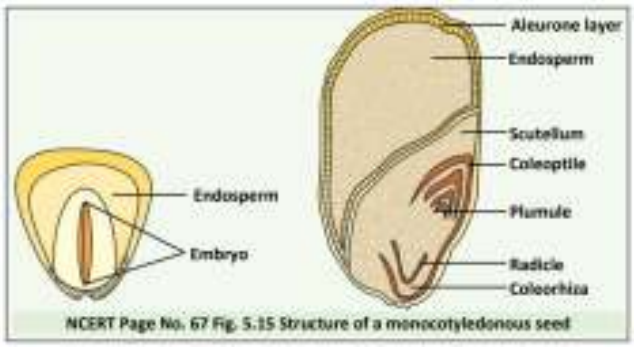


Final Answer : (1)

107) **Explain Question:** The question asks about the phenomenon by which the Ovary develops into fruit without **fertilization.**
Concept: Fruit.
Solution: The phenomenon where in the ovary develops into a fruit without fertilisation is called Parthenocarpy.
So, the correct answer is: 1. Parthenocarpy.
Final Answer: (1)

108) **Solution:** A few millimeters above the root cap is the region of meristematic activity. Cells are very thin walled with dense protoplasm and divide repeatedly.
Final Answer : (3)

109) **Solution :** A - Endosperm; B - Aleurone; C - Plumule; D - Seed coat and fruit wall



Final Answer : (4)

110) **Solution :** In this type of inflorescence, the main axis/peduncle Cymose inflorescence terminates in a flower hence is limited or definite in growth. Flowers are arranged in basipetal succession/ basipetal order. e.g. *Solanum*
Final Answer : (4)

111) **Explain Question:** The question asks about the meaning of the symbols G with a line below and G with a line above used in floral formula.
Concept: Semi technical description of flower
Solution: G̲ (underline below G): Indicates a superior ovary (hypogynous flower).
G (line above G): Indicates an inferior ovary (epigynous flower).
The symbols directly represent the position of the ovary with respect to other floral parts.
Final answer : (2)

112)

Explain Question : The question asks which symbol represents the epipetalous condition, where stamens are attached to petals.

Concept - Morphology of Flowering Plants –Family Solanaceae

Solution : Epipetalous means stamens are attached to petals (e.g., Brinjal, *Datura*).

In floral formula representation, this condition is shown by $\overbrace{C_{(5)}\ A_5}$
Final answer : (1)

113)

Explain Question : The question asks to identify the family represented by the given floral diagram based on its structural features.
Concept : Morphology of Flowering Plants-Floral diagram and characteristics of Solanaceae family
Solution : The given floral diagram shows the following features:

A. Actinomorphic and bisexual flower
B. Pentamerous arrangement
C. Gamosepalous calyx
D. Gamopetalous corolla
E. Five epipetalous stamens

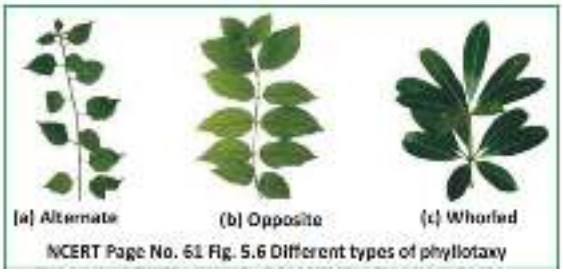
F. Bicarpellary, syncarpous ovary with axile placentation, placenta swellon

These characteristics are typical of the family Solanaceae (e.g., *Datura*, Brinjal, Chilli).
Final answer : (1)

114) **Solution :** Tomato, Brinjal, Patato, Chilli, *Datura*, Belladonna, Ashwagandha (*Withania*), Tobacco, *Petunia*, Makoi (*Solanum nigrum*) are the example of family Solanaceae & floral formula of this family is $\text{Br or Ebr} \oplus \female K_{(5)} \overset{\frown}{C_{(5)}} A_5 \underline{G_{(2)}}$
Final Answer : (1)

115) **Solution : Imbricate** : When both margins of the one petal are covered by the other two petals and both margin of another one, covers other. Rest are arranged in twisted manner. OR If the margins of sepals or petals overlap one another but not in any particular direction, then it is known as imbricate aestivation. e.g. *Cassia*, Gulmohur (*Delonix regia*).
Final Answer : (2)

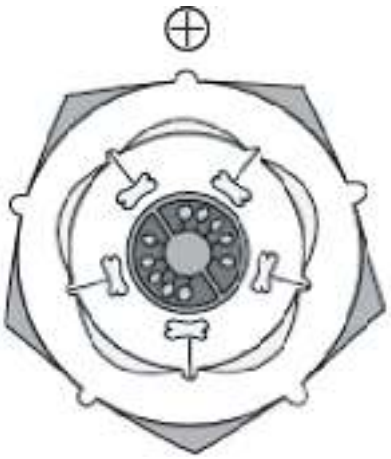
116) **Solution :** A = Alternate, B = Opposite, C = Whorled



Final Answer : (3)

117)

Question Explanation: The question is asking to identify the correct floral description based on the diagram.



Concept: Floral characters (aestivation & placentation)
Solution: The given floral diagram is of the Solanaceae family which shows united sepals and petals with valvate aestivation.
Stamens are epipetalous (attached to petals). Ovary is bilocular with axile placentation, typical

of such flowers.
Final Answer : (3)

118) **Explain Question :** Question asks to identify (A) the type of placentation in which placenta develops at the base of ovary with a single ovule, and (B) & (C) the plant families in which it is found.
Concept : Morphology of Flowering Plants -Flower : Types of placentation
Solution : Basal : In this type of placentation a single ovule is attached at the base of ovary. e.g. Marigold, Sunflower (Asteraceae/ Composite family), Gramineae / Poaceae family.
Final Answer : (2) A : Basal, B : Asteraceae, C : Poaceae

119)

Explain Question – The question asks how many statements about the family Compositae (Asteraceae) are correct.
Concept -Morphology of Flowering Plants- Inflorescence
Solution – Most advanced type of inflorescence – Correct.
Gynoecium has superior ovary – Incorrect; ovary is inferior.
Disc florets are unisexual female flower – Incorrect; disc florets are usually bisexual.
This is a type of racemose inflorescence – Correct.
Final answer : (1)

120) **Solution:** A-ii, B-iii, C-i
A. Actinomorphic — (ii) Tomato
B. Zygomorphic — (iii) Pea
C. Asymmetrical — (i) *Canna*
Final Answer : (3)

121)

Explain the Question : The question asks how many of the given statements related to vascular bundles and cambium are correct
Concept : This question is based on anatomy of stem and secondary growth
Solution :
(A) Vascular bundles of monocot stem are surrounded by sclerenchymatous bundle sheath, but this is not true for leaves → False
(B) Vascular cambium is involved in secondary growth, not primary growth → False
(C) Ring arrangement of vascular bundles is a characteristic feature of dicot stem → True
(D) Intrafascicular cambium is present within a vascular bundle → True
Thus, only two statements (C and D) are true.
Final answer : (4)

122) **Solution :** Sieve cells, albuminous cells, phloem parenchyma & phloem fibres found in gymnosperms. Albuminous cells are special type of cells are attached with the sieve cells in gymnosperm and in pteridophytes in place of companion cells.
Final Answer : (1)

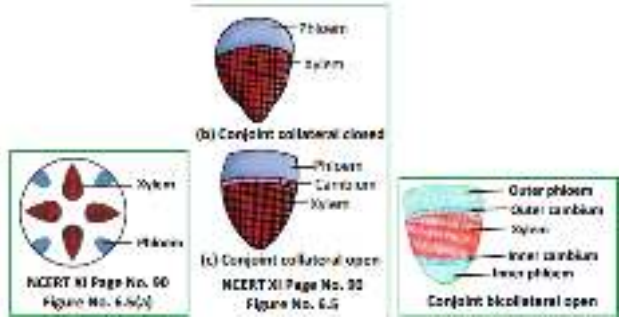
123)

The correct answer is:
3.C,A,B, D
Explanation:
In a young dicot stem, the components are arranged in the following order from outer to inner side:

A. **C (Epidermis):** The outermost layer of the stem, protecting the internal tissues.
B. **A (Cortex):** The region just beneath the epidermis, consisting of parenchyma, collenchyma, or sclerenchyma cells, providing support and storage.
C. **B (Pericycle):** A layer of cells just inside the cortex, responsible for the formation of lateral roots and contributing to secondary growth.
D. **D (Pith):** The central region of the stem, made up of soft, spongy parenchyma cells that store and transport nutrients.

Thus, the correct order from outer to inner is **C, A, B, D.**

124) **Solution :**



Final Answer : (3)

125) **Solution :** Plant and animals both show the structural similarities and variations in their external morphology as well as in their internal structure (Anatomy).
Growth in plants is largely restricted to specialised regions of active cell division called meristems.
Intercalary Meristem : It is an example of primary meristem. Intercalary meristem occurs between mature tissues. They occur in grasses and regenerate parts removed by the grazing herbivores.
Final Answer : (2)

126) **Solution :** During the formation of leaves and elongation of stem, some cells "Left behind" from shoot apical meristem, constitute the axillary bud. Such buds are present in the axils of leaves and are capable of forming a branch or a flower.

Intercalary Meristem : It is an example of primary meristem. Intercalary meristem occurs between mature tissues. They occur in grasses and regenerate parts removed by the grazing herbivores.

Apical Meristem : The meristems which occur at the tips of roots and shoots and produce primary tissues are called apical meristems. **Collenchyma-** Collenchyma is a living mechanical tissue. It devoid of lignin. It is found in the stems of herbaceous dicotyledons (young dicot stem) below the epidermis either as a homogenous layer (in sunflower stem) or in

patches (in Cucurbita stem).

Final Answer : (4)

127) **Explain Question :** Question asking about the functions of sieve tubes in phloem.
Concept : This question is based on Complex tissue -phloem
Solution : The functions of sieve tubes are controlled by nucleus of Companion cells in phloem.
Final Answer : (2)

128) **Explain Question :** Question asking about monocot stem, in question given characteristic feature of monocot stem.
Concept : This question is based on anatomy of monocot stem
Solution : Given characters are related to monocot stem. In monocot stem a large number of scattered vascular bundles, each surrounded by a sclerenchymatous bundle sheath.
Example of monocot stem - Grass
Final Answer : (3)

129) **Solution :** A–iv, B–iii, C–ii, D–i

A. In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells or motor cells . When the bulliform cells in the leaves have absorbed water and are turgid, the leaf surface exposed. When they are flaccid due to water stress, they make the leaves curl inwards to minimise water loss.
B. Stomata regulate the process of transpiration and gaseous exchange. Each stoma is composed of two bean shaped cells known as guard cells which enclose stomatal pore.
C. Thin walled, rounded, colourless, parenchymatous cells are called complementary cells. These cells are not suberized. As the complementary cells increase in number, pressure is exerted on the epidermis due to which it ruptures, forming a lens-shaped openings called lenticels.
D. In dicot leaf, mesophyll is differentiated into palisade parenchyma (palisade mesophyll or palisade tissue) and spongy parenchyma (spongy mesophyll or spongy tissue).

Final Answer : (4)

130) **Solution :** Meristematic tissues (undifferentiated) - Cells have capacity of division , Composed of immature cells.The parenchyma performs various functions like storage, photosynthesis (chlorenchyma), secretion, buoyancy (Aerenchyma).Collenchyma is a living mechanical tissue.
Types of sclerenchyma :- On the basis of variation in form, structure, origin & development sclerenchyma cells are of two types.
(i) Sclereids, (ii) Sclerenchymatous fibres.
Final Answer : (1)

131) **Solution :** Due to presence of root cap, position of root apical meristem is sub terminal/sub apical maximum growth in root takes place behind the apex.
Final Answer : (1)

132) **Solution :** Internal structure of a typical **Monocotyledonous stem** shows following anatomical features :-

A. Each vascular bundle is conjoint, collateral and closed and xylem is endarch. Each vascular bundle is surrounded by sclerenchymatous bundle sheath. Water containing cavities are present within the vascular bundles.
B. **Hypodermis** :- Hypodermis of monocotyledonous stem is made up of sclerenchyma. It is 2-3 layered thick. In monocot stem rigidity is more in hypodermis whereas in dicot stem elasticity is more. It provides mechanical support to the plant.
C. **Phloem** :- It consists of sieve tube elements and companion cells. Phloem parenchyma is absent.

Final Answer : (1)

133) **Solution :** Normally secondary growth takes place in root & stem of dicotyledons 8 gymnosperms.
→ Secondary growth is not found in leaves and monocots.
Final Answer : (3)

134)

Explain Question - The question asks to identify the incorrect statement related to plant anatomy and meristematic tissues.

Concept - Anatomy of flowering plants -complete

Solution -
Statement 1 is correct: Axillary buds arise from the shoot apical meristem.
Statement 2 is correct: In sunflower (dicot) stem, sclerenchymatous pericycle occurs above phloem in semi-lunar patches.
Statement 3 is incorrect: In a conjoint vascular bundle, phloem is located on the outer side of xylem, not inner.
Statement 4 is correct: Phloem parenchyma is absent in monocot stems .

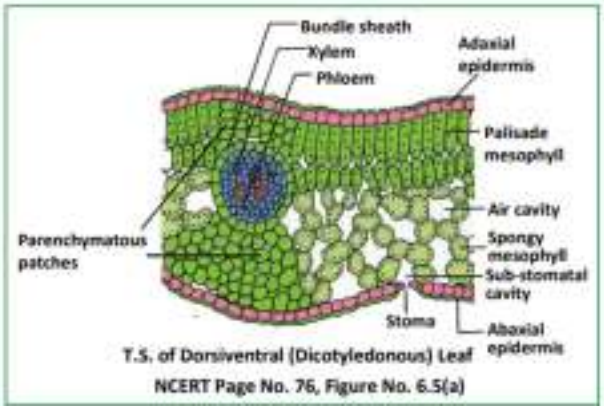
Final answer - Option 3

135) **Solution :** Sometimes, a few epidermal cells, in the vicinity of the guard cells become specialised in their shape and size and are known as subsidiary cells. The cells of the epidermis of dicot stem produce hairs called trichomes.
Final Answer : (3)

136) **Explain Question :** Question asking select incorrect statement about given diagram (given diagram is of collenchyma tissue)
Concept : This question is based on Collenchyma
Solution : The given diagram represents collenchyma tissue.
(i) Living mechanical tissue-**Correct**
(ii) Play significant role in pulp formation. **Incorrect for collenchyma**
Parenchyma is play significant role in pulp formation
(iii) Wall becomes thick due to deposition of lignin and suberin- **Incorrect for collenchyma**
Wall becomes thick due to deposition of pectin
(iv) Present in petiole of leaf. **Correct**
Final Answer : (2)

137) **Explain Question :** Question asking about function of parenchyma
Concept : This question is based on Parenchyma
Solution : Parenchyma performs various functions like photosynthesis, storage, secretion etc.
Final Answer : (2)

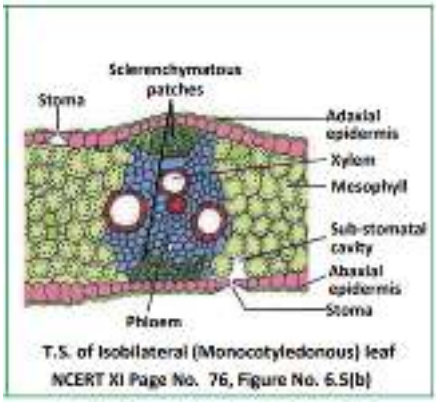
138) **Solution :**



Final Answer : (2)

139)

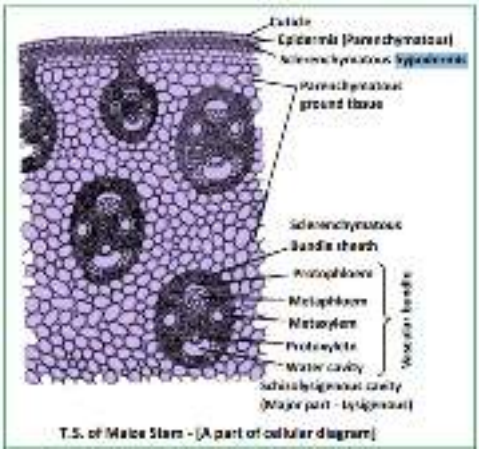
Solution :



Final Answer : (2)

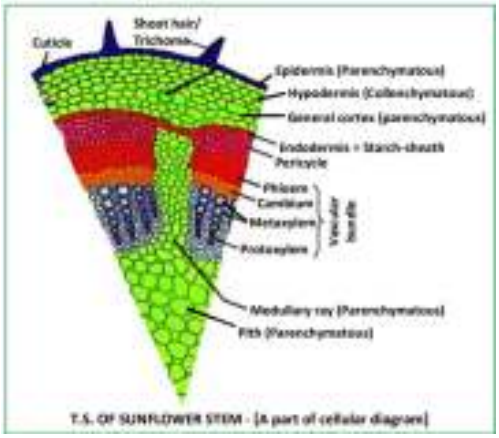
140) **Solution :** In grasses, certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells or motor cells . When the bulliform cells in the leaves have absorbed water and are turgid, the leaf surface exposed. When they are flaccid due to water stress, they make the leaves curl inwards to minimise water loss.
Final Answer : (2)

141) **Solution :**



Final Answer : (4)

142) **Solution :**



Final Answer : (2)

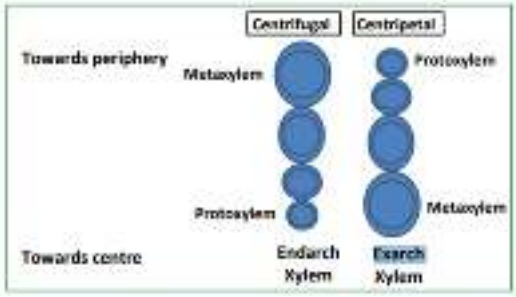
143) **Solution :** The companion cells help in maintaining the pressure gradient in the sieve tubes.
Phloem fibres are generally not found in primary phloem but are found in the secondary phloem.
Companion cells are found only in angiosperms, Sieve tube elements, companion cells, phloem parenchyma & phloem fibres (In angiosperms).
Final Answer : (4)

144) **Solution :** In Monocotyledonous stem many vascular bundles are found scattered in the ground tissue and V.B. are generally oval (egg shaped). Peripheral vascular bundles are generally smaller than the centrally located ones.
Final Answer : (4)

145)

Solution : In dicotyledonous root the number of xylem patches and phloem patches are usually two to four but they may be two to six (diarch to hexarch). Number of xylem bundles are usually more than six (polyarch) in monocotyledonous root. Normally secondary growth takes place in roots and stems of dicotyledons & gymnosperms.
Final Answer : (4)

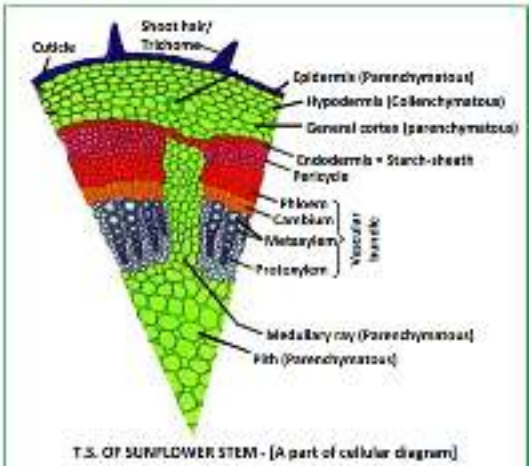
146) **Solution :** Protoxylem is formed towards the periphery and metaxylem is formed towards the centre (pith). In this condition xylem is called exarch. ex. Roots.



Final Answer : (3)

147)

Solution :



Final Answer : (4)

148) **Solution :** Sclereids are commonly found in fruit walls of nuts, pulp of guava, pear & sapota; seed coats of legumes and leaves of tea etc.
Final Answer : (2)

149) **Explain Question :**The question asks to identify the labels (A), (B), (C), and (D) shown in the four given diagrams
Concept : Tissue System → Epidermal Tissue System, Types of Vascular Bundles.
Solution : (A) = Guard cells; (B) = Radial condition; (C) = Conjoint closed; (D) = Conjoint open
Final Answer : (4)

150)

Question Explanation: Describe the structure of root hairs.
Concept: Plant Anatomy / Root System
Solution: Root hairs are delicate, thread-like structures that extend from the surface of a plant's root. They are not made of multiple cells; instead, each root hair is a tubular outgrowth or elongation of a **single epidermal cell**. Their primary function is to dramatically increase the surface area for water and nutrient absorption from the soil.
Final Answer: Unicellular elongation of epidermal cells

151) **Solution :** Compound eyes provide mosaic vision for detecting movement; antennae are long, filiform and sensory for touch, smell and air currents. Simple eyes (ocelli) are absent in cockroach.
Final Answer : (2)

152) **Solution :** The mandibles are stout and toothed for cutting food, maxillae and labium help in manipulation, suiting an omnivorous diet.
Final Answer : (2)

153) **Solution :** Tegmina are thick, leathery and cover the membranous hindwings at rest, providing protection; they are not used for active flight.
Final Answer : (2)

154) **Solution :** Abdominal segments are 10 in cockroach. In females 7th sternum is boat shaped and together with 8th and 9th sterna forms a genital pouch.
Final Answer : (3)

155) **Solution :** A pair of short, thread like and styles present in male cockroach. Anal cerci present in male and female cockroach.
Final Answer : (1)

156) **Solution :** Gizzard is a muscular grinding organ of foregut lined with chitinous teeth that mechanically breaks food into fine particles before it reaches the midgut
Final Answer : (2)

157) **Solution :** 100–150 Malpighian tubules arise at the midgut–hindgut junction and function as excretory organs.
Final Answer : (2)

158) **Solution :** Two thoracic + eight abdominal spiracles regulate entry of air into the tracheal system.
Final Answer : (2)

159) **Solution :** Mushroom gland (utriculi majores) and conglobate gland secrete seminal fluid and form spermatophore.
Final Answer : (2)

160) **Solution :** In female cockroach the ootheca is a dark reddish to blackish brown capsule total 14 to 16 ootheca are produced by female cockroach.
Final Answer : (3)

161) **Solution :** Paurometabolous means there is a development through a nymphal stage
Final Answer : (2)

162) **Solution :** : Paired salivary glands with reservoirs secrete saliva containing amylase that starts carbohydrate digestion in the buccal cavity.
Final Answer : (3)

163) **Solution :** Crop stores food; gizzard grinds it; rectal papillae reabsorb water from faeces; hepatic caeca secrete a mixture of enzymes, not cellulase alone.
Final Answer : (2)

164) **Solution :** Uric acid formation and excretion via Malpighian tubules is an adaptation to conserve body water in terrestrial life
Final Answer : (1)

165) **Solution :** Standard numbers given in the chapter match perfectly.
Final Answer : (1)

166) **Solution :** In the diagram, the gizzard appears as a barrel-shaped structure posterior to the crop with six internal folds bearing chitinous teeth for grinding food particles. This mechanical breakdown is essential before food enters the midgut for enzymatic digestion.
Final Answer : (2)

167) **Solution :** No distinct neck or tail is present; limbs are attached directly to the trunk, making the body compact for jumping and swimming.
Final Answer : (2)

168) **Solution :** Aestivation protects the frog from extreme summer heat and desiccation by burying in mud; hibernation occurs in winter for cold protection.
Final Answer : (2)

169) **Solution :** Liver produces bile (stored in gall bladder) for emulsification of fats; pancreas secretes digestive enzymes.
Final Answer : (2)

170) **Solution :** A-ii, B-i, C-iii, D-iv

Column-I	Column-II
A. Tracheoles	ii. Site of gas exchange in cockroach
B. Spiracles	i. Openings on cockroach body
C. Cutaneous respiration	iii. Respiration via skin in frog
D. Pulmonary respiration	iv. Respiration via lungs in frog

Final Answer : (1)

171) **Solution :** a - Uric acid, b - Urea, c - Ammonia

Final Answer : (2)

172) **Solution :** Testes connect to kidney via vasa efferentia; Bidder’s canal in kidney carries sperms into the urinogenital duct.
Final Answer : (2)

173) **Solution :** Frogs have bulged eyes. There eye lids are present upper, lower and nictitating membrane.
Final Answer : (1)

174) **Solution :** : Lining of buccal cavity and pharynx supplements respiration when mouth is kept closed on land.
Final Answer : (2)

175) **Solution :** : It is a digestive gland pouring juice into intestine for protein, carbohydrate and fat digestion.
Final Answer : (2)

176) **Solution :** : Frog is dioecious (unisexual) but sexes are similar externally except during breeding (males develop vocal sacs and nuptial pads).
Final Answer : (3)

177) **Solution :** : Gills are only in tadpole stage; adult uses skin, lungs and buccal lining efficiently.
Final Answer : (2)

178) **Solution :** Single ventricle receives blood from both atria, causing partial mixing – adaptation for amphibian physiology.
Final Answer : (1)

179) **Solution :** Heart has two atria + one ventricle; Bidder’s canal in males; nictitating membrane protects eye; cloaca is common chamber.
Final Answer : (2)

180) **Solution :** Liver is the largest gland; it secretes bile (stored in gall bladder shown nearby) that emulsifies fats in the intestine. The diagram clearly shows its position above the stomach.
Final Answer : (2)